CAPTESSAR AVIATION

AIRLINES PREPARATION & ATPL ORAL EXAMS NOTES

By Capt. S. R. MEENA

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For Both Papers

For General Nav Only

For Radio & Insts Only

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ACARS AND ITS CONTINUOUS OPERATION DURING FLIGHT

*Aircraft Communications Addressing and Reporting System (ACARS) a digital data link system, is used for transmission of short messages between aircraft and ground stations via air band radio (HF and VHF) or satellite and also for flight tracking.

Major functions of ACARS are as follows:

a. It automatically detects and report changes to the flight phases.
b. Interfaces with FMS for communication of flight plans and weather information from the ground station.
c. Transmits information from the aircraft to ground stations about the conditions of various aircraft systems and sensors on real-time basis including maintenance faults.

ACARS data is, therefore, of great significance in locating the last position of the aircraft and launch search and rescue after it has met with an accident.

INSTRUCTIONS FOR COMPLIANCE:

- Aircraft wherein the ACARS system is not available/disabled, operator must ensure real time flight tracking using ADS-B.
- Operators must ensure that ACARS/ADS-B are fully functional before every departure.
- Strict instructions should be given to the flight crew not to switch it off during the flight.
- Areas where there is no coverage of ACARS/ADS-B, operator should devise a procedure for effective tracking of the aircraft.
- While flying over such areas, the flight crew should report the aircraft coordinates, speed and altitude at an interval of not exceeding 15 minutes.

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ALL WEATHER OPERATIONS (AWO)

Purpose of AOM: Aerodrome Operating Minima are established in order to ensure the desired level of safety in Aeroplane Operations at an Aerodrome by limiting these operations in specified weather conditions.

Aerodrome operating minima:
The limits of usability of an aerodrome for:

a. take-off, expressed in terms of RVR and / or visibility and, if necessary, cloud conditions;
b. landing in 2Dinstrument approach operations, expressed in terms of visibility and/or RVR; ***MDA /height*** and, if necessary, cloud conditions; and
c. **landing in 3D** instrument approach operations, expressed in terms of **visibility and/or RVR** and **DA/height** appropriate to the type and/or category of the operation.

***Alert height:** A height above the runway threshold based on the characteristics of the aeroplane and its fail operational landing system, above which Cat-III operation would be discontinued and a missed approach initiated if a failure occurred in one of the redundant parts of the landing system, or in the relevant ground equipment.

All weather operations: Any surface movement, take-off and departure, approach or landing operations in conditions where visual reference is limited by weather conditions.

***Approach ban point:** The point on a final approach where the reported weather conditions at the runway must meet the applicable minima so as to be able to meet regulatory requirements for continuing an instrument approach to a landing.

***Categories of aeroplanes**: Categories of typical aeroplanes have been established based on the indicated airspeed at threshold (V_{at}) which is equal to the stall speed V_{so} multiplied by 1.3 or stall speed V_{s1g} multiplied by 1.23 (whichever is higher) in the landing configuration at maximum certificated landing mass.
Category A — less than 91 kt IAS
Category B — 91 kt or more but less than 121 kt IAS
Category C — 121 kt or more but less than 141 kt IAS
Category D —141 kt or more but less than 166 kt IAS
Category E — 166 kt or more but less than 211 kt IAS

Ceiling: The height above the ground or water of the base of the lowest layer of cloud below 6000 m (20 000 ft) covering more than half the sky.

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Circling approach: An extension of an instrument approach procedure which provides for visual circling of the aerodrome prior to landing.

***Converted Meteorological Visibility (CMV):** A value equivalent to an RVR which is derived from the reported meteorological visibility, as converted in accordance with the specified requirements in the CAR

Decision altitude (DA) or decision height (DH):

A specified altitude or height in a3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Fail-operational automatic landing system: An automatic landing system is fail operational if, in the event of a failure, the approach, flare and landing can be completed by the remaining part of the automatic system.

Final approach: That part of an instrument approach procedure which commences at the specified final approach fix or point, or where such a fix or point is not specified,

a. at the end of the last procedure turn, base turn or inbound turn of a race track procedure, if specified; or
b. At the point of interception of the last track specified in the approach procedure; and ends at a point in the vicinity of an aerodrome from which:

1. A landing can be made; or
2. A missed approach procedure is initiated.

Final approach segment (FAS): That segment of an instrument approach procedure in which alignment and descent for landing are accomplished.

Flight visibility: The visibility forward from the cockpit of an aircraft in flight.

GLS: An instrument approach operation that is based on GBAS.

Ground-based augmentation system (GBAS): An augmentation system in which the user receives augmentation information directly from a ground-based transmitter.

Ground visibility: The visibility at an aerodrome as reported by an accredited observer or by automatic systems.

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***ILS critical area:** An area of defined dimensions where vehicles, including aircraft, are excluded during all ILS operations. The critical area is protected because the presence of vehicles and/or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

***ILS sensitive area**: An area extending beyond the critical area where the parking and/or movement of vehicles, including aircraft, is controlled to prevent interference to the ILS signal during ILS operations.

Instrument approach operations:

An approach and landing using instruments for navigation guidance based on an instrument approach procedure.

a. a 2D instrument approach operation, using lateral navigation guidance only; and

b. a3D instrument approach operation, using both lateral and vertical navigation guidance.

Instrument approach operations:

Type A: a MDH or DH at or above 75 m (250 ft); and
Type B: a decision height below 75 m (250 ft).

Type B instrument approach operations are categorized as:

Category I: a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a RVR not less than 550 m;

Category II: a DH lower than 60 m (200 ft), but not lower than 30 m (100 ft) and a RVR not less than 300 m;

Category IIIA: a decision height lower than 30 m (100 ft) or no decision height and a runway visual range not less than 175 m;

Category IIIB: a decision height lower than 15 m (50 ft), or no decision height and a RVR less than 175 m but not less than 50 m; and
Category IIIC: no decision height and no RVR limitations.

Non-precision approach (NPA) procedure: An instrument approach procedure designed for 2D instrument approach operations Type A.

Precision approach (PA) procedure: An instrument approach procedure based on navigation systems (ILS, MLS, GLS and SBAS Cat I) designed for 3D instrument approach operations Type A or B.

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Instrument meteorological conditions (IMC): Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling*, less than the minima specified for visual meteorological conditions.

Low visibility procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Low visibility take-off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400 m.

MDA/MDH: A specified altitude or height in a 2 D instrument approach or circling approach/ NPA below which descent must not be made without the required visual reference.

Missed approach point (MAPt): That point in an instrument approach procedure at or before which the prescribed missed approach procedure must be initiated (in order to ensure that the minimum obstacle clearance is not infringed).

Procedure turn: A manoeuvre in which a turn is made away from a designated track followed by a turn in the opposite direction to permit the aircraft to intercept and proceed along the reciprocal of the designated track.

Required Navigation Performance (RNP): A statement of the navigation performance necessary for operation within a defined airspace.

***Runway holding position:** A designated position intended to protect a runway, or an ILS critical/sensitive area at which taxiing aircraft and vehicles shall stop and hold, unless otherwise authorised by the aerodrome control tower.

***Runway visual range (RVR):** The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

Touchdown zone (TDZ): The portion of a runway, beyond the threshold, where it is intended landing aeroplanes first contact the runway.

Vertical Navigation (VNAV): A method of navigation which permits aircraft operation on a vertical flight profile using altimetry sources, external flight path references, or a combination of these.

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Visibility: Visibility for aeronautical purposes is the greater of:

a) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b) The greatest distance at which lights in the vicinity of 1 000 candelas can be seen and identified against an unlit background.

Visual meteorological conditions (VMC): Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling*, equal to or better than specified minima.

AERODROME OPERATING MINIMA (AOM) - GENERAL

There are two sets of Aerodrome Operating Minima for application by Indian operators and at Indian aerodromes;

Normal Aerodrome Operating Minima and
Restricted Aerodrome Operating Minima.

*Normal AOM is to be applied by scheduled operators.
Restricted AOM consists of additives of height and visibility to the normal AOM and is to be applied in the following cases;

a. By non-scheduled and general aviation operators who have not been authorised normal AOM.
b. By PICs till they have gained command experience of 100 hours on type.

*Commencement and Continuation of Approach (Approach Ban Policy):

1. The PIC shall not commence an instrument approach if the reported RVR/Visibility is below the applicable minimum.
2. *If, after commencing an instrument approach, the reported RVR/Visibility Falls below the applicable minimum, the approach shall not be continued:

a. below 1 000 ft above the aerodrome; or
b. into the final approach segment.

3. *Where the RVR is not available, RVR values may be derived by converting the reported visibility.
4. *If, after entering the final approach segment or descending below 1000 ft above the aerodrome elevation, the reported RVR/visibility falls below the applicable minimum, the approach may be continued to DA/H or MDA/H.

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5. The approach may be continued below DA/H or MDA/H and the landing maybe completed provided that the required visual reference is established at the DA/H or MDA/H and is maintained.
6. *The touch-down zone RVR is always controlling.
7. *If reported and relevant, the mid-point and stop-end RVR are also controlling.

Non-precision Approaches:

*All non-precision approaches shall be flown using the Continuous Descent Final Approaches (CDFA) technique unless otherwise approved by the DGCA for a particular approach to a particular runway.

When calculating the minima the operator shall ensure that the applicable minimum **RVR is increased by200m for Cat A/B aeroplanes** and by **400m for Cat C/D aeroplanes** for approaches not flown using the CDFA technique, providing that the resulting RVR/CMV value does not exceed 5000m.

Conversion of Reported Meteorological Visibility to RVR/CMV

An Operator should ensure that a meteorological visibility to RVR conversion is not used;

a. for takeoff,
b. for calculating any other required RVR minimum less than 800 m,
c. for visual/circling approaches,
d. or when reported RVR is available.

ble 1: Conversion of Meteorological visibility to RVR

Lighting elements in operation	RVR = Reported Meteorological. Visibility x	
	Day	Night
HI approach and runway lighting	1·5	2·0
Any type of lighting installation other than above	1·0	1·5
No lighting	1·0	Not applicable

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LOW VISIBILITY OPERATIONS - GENERAL

An operator shall not conduct Category II or III operations unless:

a. Each aeroplane concerned is certified for operations with decision heights below 200 ft, or no decision height, and equipped in accordance with CAR.

b. The operations are approved by DGCA in accordance with CAR.

c. A suitable system for recording approach and/or automatic landing success and failure is established.

d. The flight crew consists of at least two pilots; and

e. Decision height is determined by means of a radio altimeter.

*An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA.

*Scheduled operators may be authorized LVTO minima of up to 125 m. This requires that a 90 m visual segment shall be available from the cockpit at the start of the take-off run.

*Non-scheduled and general aviation operators shall not conduct take-offs below500 m RVR.

*An operator shall not use an aerodrome for Category II or III operations unless the aerodrome is approved for such operations.

VFR OPERATING MINIMA

*Special VFR flights are not permitted for commercial air transport aeroplanes.

Altitude Band	Airspace Class	Flight Visbility	Distance from Cloud
At or above 3050 m (10000 ft) AMSL	A[3] B C D E F G	8 km	1500 m horizontally 300 m (1000 ft) vertically
Below 3050 m (10000 ft) and above 900 m (3000 ft) AMSL, or above 300 m (1000 ft) above terrain, whichever is the higher	A[3]B C D E F G	5 km	1500 m horizontally 300 m (1000 ft) vertically
At or below 900 m (3000 ft) AMSL, or 300 m (1000 ft) above terrain, whichever is the higher	A[3]B C D E	5 km	1500 m horizontally 300 m (1000 ft) vertically
	F G	5 km[2]	Clear of clouds and with the surface in sight

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AERODROME OPERATING MINIMA - LANDING

Normal Aerodrome Operating Minima:

Normal AOM is applicable to **Non-Scheduled and General Aviation Operators** may subject to compliance with the following conditions;

a. The PIC and Co-pilot shall have minimum **200 hours' experience on type**.
b. The PIC and Co-pilot shall have minimum **10 hours' experience on type** in the preceding 90 days.
c. The operator shall provide adequate training including training on the relevant simulator (minimum Level C) to its pilots.

Restricted Aerodrome Operating Minima:

Restricted AOM shall be based on additives applied to the Normal AOM as below;

Restricted AOM:

Normal AOM DA (H) / MDA (H) + 100 ft and
Normal AOM Visibility/RVR + 400 m

VISUAL APPROACH

For a visual approach, an operator shall use higher of the associated non-precision approach minima or **minimum visibility/RVR of 2800 m for Category A/B** aeroplanes, **3200 m for Category C aeroplanes and 3600 m for Category D** aeroplanes.

If visual approach is requested for a runway which has only a circling approach, the ground visibility shall not be less than 5 Km.

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CANPA

NPAs (in CANPA) were designed **with and without step-down fixes** in the final approach segment. **Step-downs** flown without a constant descent will require multiple thrust, pitch, and altitude adjustments inside FAF. These adjustments **increase pilot workload and potential errors** during a critical phase of flight.

NPAs designed without stepdown fixes in the final segment allow pilots to immediately descend to the MDA after crossing the FAF.

In both cases, the aircraft remains at the MDA until descending for the runway or reaching the missed approach point (MAP).

CANPA

The idea of CANPA is to fly a constant angle approach even if a glide slope or visual slope indicator such as a VASI or PAPI is unavailable.

CANPA involves making a stabilized constant angle descent rather than a quick descent to the MDA followed by flying level at the MDA until the runway environment is in sight.

In CANPA upon reaching the MDA, a decision shall be made to continue with a constant angle or level off till MAP depending on the visual condition.

In CDFA it shall be emphasize upon approaching the MDA, only two options exist for the crew:

1. Continue descend below MDA to land with the required visual reference.
2. Execute a missed approach. There is no level flight segment after reaching the MDA).

Stabilized Approaches:

*A stabilized approach is a key feature to a safe approach and landing. Operators are required by the DGCA to use the stabilized approach concept **to help eliminate CFIT**.

The **stabilized approach** concept is characterized by maintaining a stable approach speed, descent rate, vertical flight path, and configuration to the landing touchdown point.

An aeroplane should depart the FAF configured for landing and on the proper approach speed, power setting, and flight path before descending below the minimum stabilized approach height.

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Continuous Descent Final Approach (CDFA):

A technique, consistent with stabilized approach procedures, for flying the final approach segment of an NPA as a continuous descent, **without level-off**, from an altitude/height at or above the final approach fix to a point approximately 15 m (50 ft) above the landing runway threshold.

*Advantages of CDFA:

CDFA offers the following advantages:
(a) Increased safety by employing the concepts of stabilized approach.
(b) Improved pilot situational awareness and reduced pilot workload.
(c) Improved fuel efficiency by minimizing the low-altitude level flight time.
(d) Reduced noise level by minimizing the level flight time at high thrust settings.
(e) Reduced CFIT

VDA Design:

The VDA is calculated from the Final Approach Fix (FAF) altitude to the threshold crossing height (TCH).

The optimum NPA descent angle (VDA) is 3.0 degrees.

Although VDA should not exceed 4.5 degrees for Cat A and B aeroplanes or 3.77 degrees for Cat C and D aeroplanes.

*Derived Decision Altitude (DDA):

It is required that flight crews add a prescribed altitude increment to the MDA depending on type of aircraft, (minimally 50 feet) to determine the altitude at which the vertical transition to the missed approach should be initiated in order to prevent descent below the MDA.

Visibility Minima Penalty:

A visibility penalty of **200 m for Cat A and B**, 400 m for Cat C and D aeroplanes will apply to the published approach minima if operators do not use CDFA on NPAs.

Q: What are the domestic routes on Jeppesen enroute chart? And which of them are only one way
Ans. HVJW are domestic routes. HV are one way routes.
Q. What is TCH?
Ans. Threshold Crossing Height. Height of the effective visual glide path over the threshold.

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AIRSPACES

Classification of airspaces

ATS airspaces shall be classified and designated in accordance with the following:

Class A. IFR flights only are permitted, all flights are provided with air traffic control service and are separated from each other.

Class B. IFR and **VFR** flights are permitted, all flights are provided with air traffic control service and are separated from each other.

Class C. IFR and VFR flights are permitted, all flights are provided with air traffic control service and **IFR flights are separated from other IFR flights and from VFR flights.**

VFR flights are separated from IFR flights and receive traffic information in respect of other VFR flights.

Class D. IFR and VFR flights are permitted and all flights are provided with air traffic control service, IFR flights are separated from other IFR flights and receive traffic information in respect of VFR flights, VFR flights receive traffic information in respect of all other flights.

Class E. IFR and VFR flights are permitted; IFR flights are provided with air traffic control service and are separated from other IFR flights.

All flights receive traffic information as far as is practical.

Class E shall not be used for control zones.

Class F. IFR and VFR flights are permitted, all participating IFR flights receive an air traffic advisory service and all flights receive flight information service if requested.

Class G. IFR and VFR flights are permitted and receive **flight information service** if requested.

Air traffic control service shall be provided:

a) to all IFR flights in airspace Classes A, B, C, D and E;

b) to all VFR flights in airspace Classes B, C and D;

c) to all special VFR flights;

e) to all aerodrome traffic at controlled aerodromes

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Oceanic Control Areas (OCAs): Over large areas of the world's Oceans e.g. the North Atlantic, traffic control has special problems. To solve the problems, the airspace **above FL55** over the Oceans is designated as OCAs where strict rules are enforced and special navigation procedures are applied.

SVFR: All classes of airspace which support the establishment of a CTR permit SVFR.

Non-radio: *Classes E-G permit VFR traffic to fly without VHF 2-way RTF communications.
Speed Limit: *A speed limit of 250 kt IAS is applied to VFR traffic in class C and all traffic in classes D - G. The limit is applicable to traffic flying below 10 000 ft (FL100) only.

Q. What ATS airspaces are available in India?
Ans. In India, ATS airspaces are designated as Class D, E, F and G and are categorized as per NOTAM G 0066/99. Delhi, Mumbai comes under Class D Airspace.

Q. What is the difference between Class D & Class E airspace in India?
Ans. In Class D Airspace, traffic information is provided for all IFR & VFR, whereas, in Class E Airspace traffic information is provided where possible.

***Q. What are the limitations for holding speeds?**
Ans. The ICAO Maximum holding speeds are defined as:

Up to 14000 ft:	230kts
14000 ft to 20000 ft:	240kts
20000 ft to 34000 ft:	265kts
Above 34000 ft:	M0.83

Q. What is the code for ILS, VOR and NDB in an approach plate?
Ans. ILS: 1 as in chart 11-1
VOR: 3 as in chart 13-1
NDB: 6 as in chart 16-1
If there are two airports in the same city like in Bangalore, then ILS will be 21-1, VOR will be 23-1 etc.

Q. When does the IAS changes from knots to Mach?
Ans. At FL 270

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CLASSES OF AIRSPACE

Class	Rules	Service	Clearance	Separation	FIS	Speed Limit (NOTE 1)	Radio Watch	CTR SVFR	Use
A	IFR	ATC	YES	ALL		NO	YES	YES	CTA: CTR AIRWAYS: ROUTES
B	IFR VFR	ATC	YES	ALL		NO	YES	YES	CTA: CTR AIRWAYS: ROUTES
C	IFR	ATC	YES	IFR-IFR IFR-VFR		NO	YES	YES	CTA: CTR (AIRWAYS) (NOTE 5)
	VFR	ATC FIS		VFR-IFR	VFR-VFR	YES			
D	IFR	ATC	YES	IFR-IFR		YES	YES	YES	CTA: CTR (AIRWAYS) (NOTE 5)
	VFR	ATC (NOTE 4)			IFR-VFR VFR-IFR VFR-VFR				
E	IFR	ATC	YES	IFR-IFR		YES	YES	NO (NOTE 2)	CTA: (AIRWAYS) (NOTE 5)
	VFR	FIS	NO		IFR-VFR VFR-IFR VFR-VFR		NO		
F	IFR	ADV ATC	NO	IFR-IFR (NOTE 3)	ALL	YES	YES	NO	ADVISORY ROUTES
	IFR VFR	FIS					YES NO		
G	IFR VFR	FIS	NO	NONE	ALL	YES	YES NO	NO	OPEN FIR; FIS ROUTES

Note 1: General Speed Limit is 250 kts IAS enforced below FL100 (10 000ft)
Note 2: Separation as far as is practical
Note 3: Service to participating IFR traffic only (non participating IFR to file FP and maintain 2-way comms)
Note 4: Radar information/advice may be provided to augment FIS in CAS
Note 5: An airway will adopt the classification of the surrounding airspace when it passes through a CTA

COMMUNICATIONS FAILURE:

In case of communication failure, the aircraft shall attempt to establish communications with the appropriate air traffic control unit using all other available means.

Where Controller-Pilot Data Link Communications (CPDLC) exists, the requirement for voice communications is required to be maintained.

Confirmation of Communication Failure:

ATC will confirm that an aircraft is subject to a communications failure by requesting it to execute a specific manouevre which can be observed by radar or transmit a signal in order to indicate acknowledgement.

In any case ATC will transmit blind instructions to the aircraft on frequencies on which the aircraft is believed to be listening.

In VMC

a. If a failure occurs whilst the aircraft is flying in VMC (regardless of flight rules - VFR or IFR), the aircraft is to continue to be flown in VMC, landed at the nearest suitable aerodrome and arrival reported by the most expeditious means to the appropriate ATCU.

b. if considered advisable, complete an IFR flight

In IMC

1. maintain the **last assigned speed and level**, or minimum flight altitude if higher, for a period of **20 minutes** following the aircraft's failure to report its position over a compulsory reporting point and thereafter adjust level and speed in accordance with the filed flight plan;
2. **in airspace where radar is used** in the provision of air traffic control, maintain the **last assigned speed and level**, or minimum flight altitude if higher, for a period of **7 minutes** following:

a. the time the last assigned level or minimum flight altitude is reached; or
b. the time the transponder is set to Code 7600; or
c. the aircraft's failure to report its position over a compulsory reporting point; whichever is later, and thereafter adjust level and speed in accordance with the filed flight plan.

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3. commence descent from the navigation aid or fix at, or as close as possible to, the expected approach time last received and acknowledged; or, if no expected approach time has been received and acknowledged, at, or as close as possible to, the estimated time of arrival resulting from the current flight plan;
4. land, if possible, within thirty minutes after the estimated time of arrival

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EDTO

***Extended diversion time operations (EDTO):** Any operation by an aeroplane with **two or more turbine engines** where the diversion time to an en-route alternate aerodrome is greater than the threshold time established by the DGCA.

The **purpose** of initial ETOPS regulations were to provide **very high level of safety** while facilitating the use of twin engines on routes, which were previously restricted(NA) to three or four engine aeroplanes.

ETOPS has now evolved to EDTO (Extended Diversion Time Operations) to encompass **two or more engine** aeroplanes and the intent of the current regulation is to avoid a diversion and if it occurs, to ensure that the diversion is safe.

*Operators shall not operate an aeroplane with two or more engines or an aeroplane of AUW more than 5700 kg beyond the threshold time unless approved by DGCA for EDTO.

EDTO Entry Point: The first point on the route of an EDTO flight; determined using a one-engine inoperative cruise speed that is more than the threshold from an en-route alternate airport for airplanes with two engines and more than two engines.

EDTO critical fuel: The fuel quantity necessary to fly to an en-route alternate aerodrome considering, at the most critical point on the route, the most limiting system failure (Engine failure, Decompression or both).

Maximum diversion time: Maximum allowable range, expressed in time, from a point on a route to an en-route alternate aerodrome.

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Threshold time: The range, expressed in time, established by the State of the Operator to an en-route alternate aerodrome, whereby any time beyond requires an EDTO approval from the State of the Operator.

1. The threshold time for EDTO established by DGCA is 60 minutes for **two engine aeroplanes** more than 45360 kgs and **scheduled operators** with two engine aeroplanes irrespective of AUW and passenger capacity.
2. The threshold time for EDTO established by DGCA is **90 minutes for Non-scheduled operators** (NSOPs) less than 45360 kgs.
3. The threshold time for EDTO established by DGCA is 120 minutes for more than two engine aeroplanes.

GENERAL REQUIREMENTS FOR EDTO:

Unless the operation has been specifically approved by DGCA an aeroplane with two or more turbine engines shall not be operated on a route where the diversion time from any point on the route, to an en-route alternate aerodrome exceeds a threshold time established for such operations by the DGCA

1. *When the diversion time exceeds the threshold time, the operation is considered to be an extended diversion time operation (EDTO).
2. *For the purpose of EDTO, the take-off and/or destination aerodromes may be considered en-route alternate aerodromes.

EDTO critical fuel:

No operator may dispatch or release for flight or takeoff a turbine engine powered airplane in EDTO unless, considering wind and other weather conditions expected, it has enough fuel to satisfy paragraphs below:

(i) The greater amount of fuel sufficient to fly to an en-route alternate under the following three scenarios:

a. Assuming a **rapid decompression** at the most critical point followed by descent to a safe altitude in compliance with the oxygen supply requirements, or

b. At the approved one-engine inoperative cruise speed assuming a **rapid decompression and a simultaneous engine failure** at the most critical point followed by descent to a safe altitude in compliance with the oxygen supply requirements, or

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c. At the approved one-engine inoperative cruise speed assuming an **engine failure at the most critical point** followed by descent to the one-engine inoperative cruise altitude.

***75/90 minutes' operation**

Approval to carry out EDTO with 75 minutes diversion time may be granted by DGCA to an operator **with minimal or no in-service experience** with particular airframe engine combination.

***More than 75/90 minutes - 120 minutes operation**

Each operator requesting approval to conduct EDTO with a maximum diversion time of 120 minutes (in still air) should have **minimum of 12 consecutive months of operational in service experience** with the specified airframe engine combination.

More than 120 minutes - 180 minutes operation

Each operator requesting approval for maximum diversion time of 180 minutes (in still air) **should have held current approval for 120 minutes, EDTO for a minimum period of 12 months** with a corresponding high level of demonstrated propulsion system reliability.

***Adequate Airfield**: In general terms an **operator** may make an appraisal that an aerodrome has long enough runways, and is sufficiently equipped, to be considered **adequate** for his planned ETOPS routes. In particular, it should be expected that at the anticipated time of use:
The aerodrome will be **available and equipped with the necessary ancillary services**, such as **ATC, sufficient lighting, communications**, weather reporting, navaids and safety cover.

***Suitable Airfield**. The **commander** must satisfy himself **on the day**, using criteria provided by the operator, that he has sufficient adequate aerodromes which, taking into account the **weather and any equipment unserviceabilities**, are suitable for his intended operation.

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FANS

Q: What is FANS?

Future Air Navigation System (FANS) is a **concept** that was developed to allow more aircraft to safely and efficiently utilize a given volume of airspace.

Today FANS is used primarily in the oceanic regions taking advantage of both satellite communication and satellite navigation to effectively create a virtual radar environment for safe passage of aircraft.

The **aim of FANS** is to provide an **integrated air traffic control system** in areas where radar is not available by using GNSS to define aircraft position with datalink and voice communications through geostationary and lower orbit satellites providing global coverage to ATC centres.

This will provide ATC with continuous information on the aircraft positions and direct pilot to controller communications. When in operation this will allow separation distances to be significantly reduced, and if position accuracy is degraded for any reason, the aircraft still have TCAS to warn of any potential conflict.

Today, **FANS-1** is the standard used on Boeing aircraft while the Airbus standard is known as **FANS-A**.

FANS A/1 provides a **communication, navigation and surveillance** (CNS) system and an **automatic dependent surveillance system** (ADS).

Currently aircraft are controlled using voice communications. **Over and close to populated landmasses**, ATC use radar to provide positive control of aircraft with **VHF communications**.

However, **over oceans, deserts and polar regions** ATC provide a procedural control service which generally requires **HF communications** and high vertical, lateral and longitudinal separation distances, resulting in a low traffic density.

Aircraft operating in the **same direction** across the **North Atlantic in MNPS and RVSM** airspace are given lateral separation of 60 nm, longitudinal separation of 10 minutes and vertical separation of 2000 ft.

In non-MNPS/ RVSM airspace the minima are 120 nm, 15 minutes and 4000 ft respectively.

Position reports are passed by aircraft crossing the North Atlantic every 10° of longitude up to 70°N and every 20° north thereof, which means ATC receive a position update every 30 – 60 minutes.

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AIRCRAFT COMMUNICATION ADDRESSING and REPORTING SYSTEM (ACARS)

ACARS uses **datalink** in telex format to pass messages between the aircraft and ATC or aircraft operating companies at the relatively slow rate of **2.4 kbps using VHF** and the messages were printed out in the cockpit.

In the early 1990s this service was extended to SATCOM for flights outside VHF coverage using geostationary satellites. The gap in polar regions was closed in 2001 by extending the service to **HF**.

VHF DATALINK (VDL)

ACARS is gradually being replaced with **VDL mode 2** which provides a digital link at a rate of 31.5 kbps, giving more than 10 times the capacity of ACARS.

VDL mode 2 requires specific address for an aircraft and the unique mode S address will be used. This type of digital system is known as **CPDLC**.

VDL mode 2 is the part of the new digital aeronautical telecommunications network (ATN) which links ATC facilities and aircraft operational control (AOC) with the aircraft.

CPDLC

***Controller-pilot data link communications (CPDLC) -** A means of communication between controller and pilot, using data link for air traffic control (ATC) communications.

CPDLC permits datalink messages to be generated for all stages of flight.

The messages have a **fixed format defined within the FMS and ATSU computers** and are activated by the ATC controller or pilot either as an instruction or response to a request.

The message formats available cover all aspects of pre-departure clearance (PDC), taxy instructions, takeoff, climb, en-route, descent and landing. Additionally, there is provision for emergency and safety messages.

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GAGAN & IRNSS

GAGAN is the acronym for **GPS Aided GEO Augmented Navigation**. The Indian Space Research Organization (ISRO) and Airports Authority of India (AAI) have implemented the **GPS Aided Geo Augmented Navigation**-GAGAN project as a Satellite Based Augmentation System (SBAS) for the Indian Airspace.

It is a system **to improve the accuracy of a global navigation satellite system (GNSS) receiver by providing reference signals.**

The **objective** of GAGAN to establish, deploy and certify SBAS for safety-of-life civil aviation applications in India has been successfully completed. The system is inter-operable with other international SBAS systems like US-WAAS, European EGNOS.

The GAGAN's **goal** is to provide a navigation system to assist aircraft in accurate landing over the Indian airspace and in the adjoining area and applicable to safety-to-life civil operations.

GAGAN provides the **additional accuracy, availability, and integrity** necessary for all phases of flight, from enroute through approach for all qualified airports within the GAGAN service volume.

This makes airline operations more efficient and effective, increase air safety, and fuel efficiency. Further, with vertical guidance at runways, a significant cost will be saved due to withdrawal of ground aids and reduced workload of airline crew and traffic controllers.

Initially, DGCA certified GAGAN for en-route operations (RNP 0.1) on December 30, 2013 and subsequently on April 21, 2015 for precision approach services (APV 1).

GAGAN is the first Satellite-Based Augmentation System in the world which has been certified for approach with vertical guidance operating in the equatorial / equatorial ionospheric region. India has become the third country in the world to have such precision approach capabilities.

TCAS

TCAS is designed to provide an additional margin of safety and keep commercial aircraft clear of conflict, independently of Air Traffic Control.

TCAS I

1. TCAS I is a **first generation** collision avoidance system and simply warns the crew of other traffic in the vicinity of their aircraft.
2. It will detect and display **range** and approximate **relative bearing**.
3. If the TCAS display aircraft and the intruder are carrying **Mode C relative altitude** will also be displayed.
4. It generates visual and aural warnings - TRAFFIC ADVISORIES (TAs): **"Traffic, Traffic"**

TCAS II

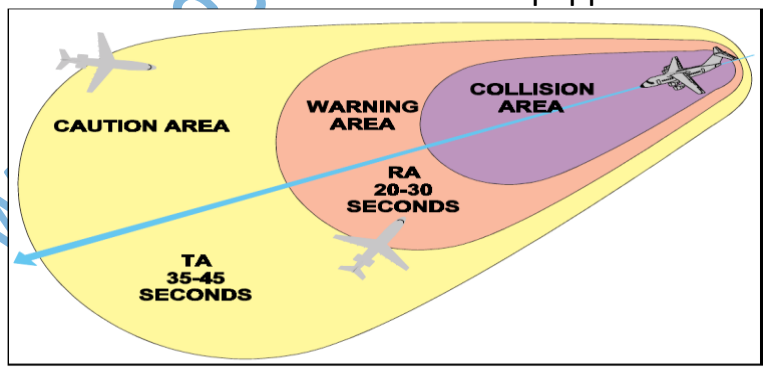
TCAS II detects intruders in the TCAS aircraft's vicinity, **assesses the collision risk and presents warnings** in the form of **TAs** and **RAs** e.g.:

"Climb" "Increase Climb" Descend" "Increase Descent" Monitor Vertical Speed"

RAs offer manoeuvring advice in the vertical plane only to resolve conflict.

PRINCIPLE

1. TCAS II operates on the secondary radar principle using the normal SSR frequencies of 1030MHz and 1090MHz, but in an air to air role.
2. Using this principle the TCAS system creates two protective three dimensional bubbles around the TCAS equipped aircraft.



AIRCRAFT EQUIPMENT

- must have a minimum of A Mode "A" Transponder.
- If the transponder is switched off, or is unserviceable, the intruding aircraft are invisible.

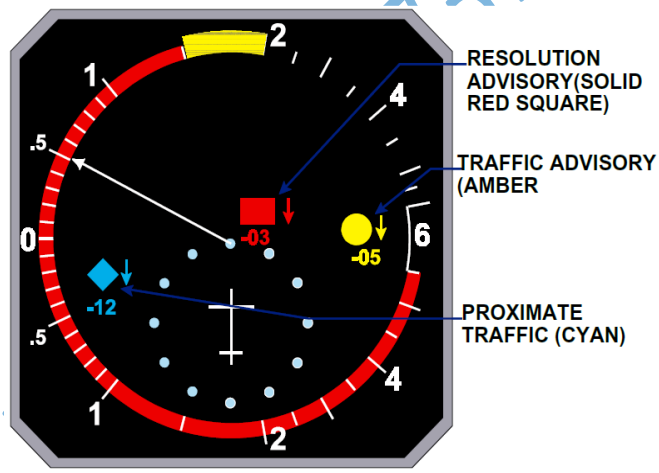
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Traffic Advisories (TAs) exist when an intruder penetrates the outer bubble caution area and is between **45 and 35 seconds** from the collision area. TA's appear as **solid amber** circles on the TCAS display.

Resolution Advisories (RAs) exist when an intruder penetrates the inner bubble warning area and is between **30 and 20 seconds** from the collision area. RA's appear as **solid red rectangles** on the TCAS display.

PROXIMATE TRAFFIC/ OTHER TRAFFIC

Proximate Traffic appears as a **solid cyan diamond** and aircraft within range of the display and within +/- 1200 feet relative height. **Other Traffic** appears as **hollow cyan diamonds** which represent transponder equipped aircraft within range of the display and within +/- 2700 feet relative height.



ACTION TO BE TAKEN:

Action on Receiving a TA: Flight crews should **commence a visual search** of that part of the sky.

Action on Receiving an RA: Pilots are to **initiate the required manoeuvre immediately**, adjusting flight path, aircraft power and trim accordingly and inform ATC as soon as possible of any deviation from an ATC clearance.

What is GPWS, and how does it work?

The GPWS is essentially a **central computer system** that receives various data inputs on configuration, height/altitude, and instrument landing system (ILS) glide slope deviation. It then calculates these inputs to detect if any of the following dangerous and/or potentially dangerous circumstances exist:

1. Excessive rate of descent

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2. Excessive terrain closure rate
3. Height loss after takeoff
4. Flaps or gear not selected for landing
5. Too low on the ILS glide slope
6. Descending below approach minima

These circumstances form the six main working modes of the GPWS.

What is EGPWS?

EGPWS provides a greater level of detection than a standard GPWS. For example, terrain mapping is a new feature on EGPWS.

It can be shown on navigation displays by using the weather system.

Probable windshear aural and visual warnings also can be generated to warn of an impending possibility of encountering windshear ahead.

What is the pilot's order of priority given a windshear, GPWS, and TCAS warning at the same time?

First: Windshear
Second: Ground proximity warning system (GPWS)
Third: Traffic collision avoidance system (TCAS)

Difference between GPWS and EGPWS

GPWS	EGPWS
No terrain database	Look ahead facility because of terrain database
In fifth mode only one GP alert	5A- Soft alert 1.3 dots 5B- Hard GP alert when 2.5 dots fly up indication
Mode 6 decision height only	6A Decision height 6B Bank angle alert
No mode 7	Mode 7 Windshear
Gives warning of the terrain below only	Gives warning of terrain below and ahead of the aircraft

EGPWS has all the capabilities of GPWS and augments it by using a Global Positioning System, or GPS, to provide very accurate information on the exact

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location of the aircraft. This is then coupled with an extensive **terrain database**; basically, a map that details how the ground changes in the area.

GPWS had one very **serious flaw;** it can only monitor the ground directly beneath it. This can be a problem if there is a very sudden change in the terrain and the GPWS cannot provide a prompt enough warning for the pilot to react to.

With EGPWS, the system can track the course of the aircraft and see if it is heading towards a mountain or other similar threat.

EGPWS is only as good as the terrain database that it has. It needs to be up-to-date and contain accurate information.

***Summary:**

1. EGPWS is safer and more advanced than GPWS
2. EGPWS uses GPS while GPWS doesn't.
3. EGPWS uses a terrain database that is not available in GPWS.
4. GPWS is only aware of the ground below it while EGPWS is aware of a larger area.

EGPWS has all the capabilities of GPWS and augments it by using a Global Positioning System, or GPS, to provide very accurate information on the exact location of the aircraft.

Q. What level of precedence should be given to warnings?

Ans. REMEMBER: SWAGAT

Stall
Windshear
GPWS
TCAS

DRMC

COMPASS REQUIREMENTS

1. **Horizontal**
2. **Sensitive**
3. **Aperiodic**

TURNING ERRORS

Turning errors are **maximum** when turning through **north** and **south**, and **zero** when turning through **east** and **west**.

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ACCELERATION ERRORS

Acceleration errors are **zero** on **N/S magnetic headings**, increasing to **maximum** on **headings 090°M and 270 ° M**.

Acceleration causes an **apparent turn** towards the **nearer pole**.

Deceleration causes an **apparent turn** towards the **further pole.**

What is a gyroscope?

A gyroscope is a body (usually a rotor/wheel) rotating freely in one or more directions that possesses the gyroscopic properties of rigidity and precession.

GIMBAL RINGS

These are the supports for the rotor of a gyroscopic instrument.

THE FUNDAMENTAL PROPERTIES OF A GYROSCOPE

1. **Rigidity.** A spinning rotor maintains its axis pointing in a fixed direction in space, unless subjected to an external force.
2. **Precession.** If an external force is applied to change the direction of the rotor axis, the gyro moves in a plane at right angles to that of the torque, the resulting movement being called **'precession**'.

FACTORS AFFECTING RIGIDITY:

The gyroscopic inertia or rigidity of a gyro rotor is directly proportional to Iω where I is the moment of inertia and ω is the angular velocity. I is mr^2.

PRECESSION RATE

The rate of precession (Ω) is proportional to torque applied and **inversely** proportional to I ω, the rigidity or gyroscopic inertia.

WANDER

1. If the axis of a gyro rotor departs from its chosen direction it is said to **wander**.
2. If the axis of the gyro rotor wanders in the horizontal plane it is said to **drift**.
3. If the rotor axis wanders in the vertical plane it is said to **topple**.

CPDLC?

Controller–pilot data link communications (CPDLC) is a method by which air traffic controllers can communicate with pilots over a datalink system

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INS

Principle- INS is the **self-contained Nav Aid on board** which senses acceleration and integrate it twice to get velocity and distance.

Measurements are provided by accelerometers and gyroscopes used to track the position and orientation of an object relative to a known position.

The **fundamental element** of this complex system is the Inertial Sensor System (ISS).

To make up this system we have a **stable platform** consisting of high quality gyros and accelerometers and a computer.

The purpose of the computer is to integrate the accelerometer outputs with time to give **velocity** and then integrate velocity with time to give **distance** travelled.

An integrating gyroscope used in INS's is a one degree of freedom gyro using viscous rather than mechanical (spring) restraint.

*ERRORS OF INS

Errors can be conveniently considered under the following headings:

1. Bounded errors
2. Unbounded errors
3. Inherent errors

IRS

The **Inertial Reference Unit** (IRU) is the heart of the **Inertial Reference System** (IRS).

THE PRIMARY SOURCES OF INFORMATION

The primary sources of information for the IRU are its own internal sensors three **laser gyros**, and three **inertial accelerometers**.

The only other inputs required are initial position, barometric altitude, and True Air Speed (TAS).

THE LASER GYRO

The **laser gyro** is an example of the application that uses the characteristics of light to measure motion.

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PRINCIPLES OF LASER GYROS AND IRS

Laser (Light Amplification and Stimulated Emission of Radiation) gyros measure rotation by comparing two laser beams created and directed to rotate in opposite directions within a very narrow tunnel.

INS	**IRS**
Use conventional gyros	Use RLG
Gyros are mounted on a pendulous platform	Mounted on a platform which is strap down with aircraft structure (unit fixture)
Less accurate	More accurate .5 nm to 2 nm/hr
Aligning time in middle latitudes is 20 min	Aligning time in middle latitudes is 3-10 min
	Longer life cycle
	Disadvantage- Laser lock, To avoid use dither motor which vibrates and opens the lock

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RNAV

An area navigation (RNAV) system is any system that **allows the aircraft to be navigated to the required level of accuracy** without the requirement to fly directly over ground based facilities.

BENEFITS OF RNAV

RNAV allows aircraft to take a more direct flight path improving the operating efficiency and helping in relieving congestion on the overcrowded airway system. Hence the benefits are:

- A reduction in distance, flight time and fuel (and hence costs).
- An increase in the present route capacity.
- A reduction in vertical and horizontal separation criteria.

TYPES AND LEVELS OF RNAV

There are two **types of RNAV:**

1. **Basic RNAV** (B-RNAV) which is required to give a position accuracy to **within 5 nm** on 95% of occasions.
2. **Precision RNAV** (P-RNAV) must be accurate to **within 1.0 nm** on 95% of occasions. P-RNAV routes are now being established in terminal airspace.

There are three **levels of RNAV** capability:

1. **2D RNAV** which relates to the capabilities in the **horizontal plane only.**
2. **3D RNAV** indicates the addition of a guidance in the **vertical plane.**
3. **4D RNAV** indicates the addition to 3D RNAV of a **timing function**.

OPERATION OF RNAV

1. **2D RNAV SYSTEM:** A simple RNAV system uses **rho/theta** to define position, which is derived from VOR/DME stations.

2. **LEVEL 4 RNAV SYSTEMS:** The RNAV in modern passenger aircraft is carried out by a flight management computer (FMC) which also provides guidance and performance functions.

The FMS provides 4D area navigation (latitude, longitude, altitude and time) and optimises performance to achieve the most economical flight possible.

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In the navigation functions the FMC receives inputs of position and heading from the IRS and fixing information using twin DME. The FMC compares these inputs and by a process known as **Kalman filtering** produces a system position.

In the operation with radio position updating, the FMC is combining the short term accuracy of the IRS with the long term accuracy of the external reference.

Strategic lateral offset procedure (SLOP)

SLOP allows aircraft to offset the centreline of an airway or flight route by a small amount, normally to the right, so that collision with opposite direction aircraft becomes unlikely.

SLOP is a solution to a byproduct of increased navigation accuracy in aircraft. If an error in height occurs, there is a much higher chance of collision.

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FMS

The FMC contains a **performance database** and a **navigation database**.

The **performance database** contains all parameters of the aircraft performance and the company's cost index strategy.

The **navigation database** contains aeronautical information for the planned area of operations of the aircraft.

The navigation data is updated every 28 days and the FMC contains the current and next 28 days Database.

PRINCIPLE OF OPERATION - TWIN IRS, TWIN FMC

In a twin IRS system, the **left FMC** will normally receive information from the **left IRS** and the right FMC from the right IRS. The FMC must have the input of an external reference in order to determine the correct position.

PRINCIPLE OF OPERATION - TRIPLE IRS, TWIN FMC

Positional information and **heading** from the triple INS/IRS is fed into the FMC where the information is compared to check for any system having gross errors and then averaged.

This position may then be compared with an **external reference**. The FMC uses **Kalman filtering** to produce position and velocity.

Kalman filtering is the process used within a navigation computer to combine the short term accuracy of the IRS with the long term accuracy of the external reference.

MINIMUM FLIGHT CREW REQUIREMENTS

1. Not less than that specified in the approved AFM or Operations Manual or CofA.
2. All aeroplanes above 1500 kgs engaged in **Scheduled / Non-Scheduled Air Transport operations**, shall be flown by at least **two** pilots.
3. All aircrafts operated by **State Governments/ VVIP/VIP** shall be flown by at least **two** pilots.
4. Aircraft above 5700 kg shall be flown by at least **two** pilots.
5. When engaged in flight **calibration, aerial survey and remote sensing work**, shall be flown by **two** pilots.
6. An aircraft having a seating capacity of **10 or more seats**, excluding pilot seat(s), shall be flown by **two** pilots.

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SINGLE PILOT OPERATIONS

Minimum Crew Requirements:

At least 100 hours Pilot-in-Command experience on the type and model of the aircraft to be flown and has met all other applicable requirements.

An aeroplane shall not be operated under the IFR by a single pilot unless:

1. the flight manual does not require a flight crew of more than one;
2. the aeroplane is propeller-driven;
3. the maximum approved passenger seating configuration is not more than nine;
4. the maximum certificated take-off mass does not exceed 5 700 kg;
5. approved by DGCA

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Long Range Flight and Polar Navigation

Q. How do you navigate to the North Pole?
Ans. Difficulties of flying near the poles are rapid change of true direction (due to convergency) and variation (due to proximity to the magnetic poles). Without an RNAV system, the solution is to ignore the compass and fly a gyro heading.
*Navigation in polar region can only be done through Grid Navigation technique. In grid navigation maps are overlaid with a grid of lines indicating gyro north to which gyroscope is aligned. While gyro steering is being employed, the magnetic element of the Gyro Magnetic Compass is disconnected.

Course and INS Cross-checking
In an emergency course and INS can be cross-checked:
In a 3 set system, the output of each system should be compared (a voting system) from which inaccuracy in any one system should be quickly detected.
In a 2 set system, the failure of one system would not be readily detected unless the system captions malfunction codes in which case interpretation of the code should reveal which unit is faulty. If it is possible to obtain a fix (from the weather radar or bearings from ADF and NDB, or a fix from a VOR beacon), comparing with the system positions should reveal the inaccurate system.

If uncertainty still exists more basic methods include contacting another aeroplane in the vicinity and cross-checking spot winds, ground speed and drift. As a last resort, comparison of the outputs from the nav systems could be compared with the flight plan data for wind velocity at the DR position of the aeroplane.

Unable to Continue in Accordance with ATC Clearance
If an aeroplane is unable to continue the flight as per the ATC clearance, a revised clearance shall, whenever possible, be obtained prior to initiating any action. This shall also apply to aircraft which are unable to maintain the specified navigation accuracy. The revised clearance shall be obtained by RTF distress or urgency traffic as appropriate.

Polar Navigation
Polar tracks defined as North/South routes involve navigation at high latitudes (above 65°N). In these areas, the lack of ground radio aids, high rates of change of magnetic variation and steep magnetic dip angles, make conventional

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airways navigation difficult if not impossible. For these reasons, **magnetic compasses become unreliable.**

Other factors:

1. **limited communications** mainly restricted to HF,
2. **lack of en-route alternate aerodromes** and
3. **high rates of gyro correction** (for earth rate and transport wander).

In this situation, navigation is achieved by reference to a **grid navigation process** or **reliance on inertial systems** and satellite based **global positioning** (GPS).

In areas where the rate of change of magnetic variation becomes excessive (in close proximity to the North Magnetic Pole), **VOR beacons are orientated to true north** to assist grid navigation.

Grid navigation in conjunction with a directional gyro can be used in polar areas to resolve polar navigation problems.

Minimum Time Routes

A minimum time route is defined as the track flown between two points which **results in the shortest time adhering to all ATC and airspace restrictions**. In reality, airspace restrictions (danger/restricted/prohibited areas), airway routings, and wind and meteorological considerations may make another longer (than GC) track a quicker option.

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MNPSA

Q. What is MNPS?

Ans. MNPS stands for Minimum Navigation Performance Specification. Due to economic, geographical and geophysical considerations, the airspace **over the North Atlantic Ocean** between the latitudes of 35N and 70N is some of the most frequently flown airspace in the World.

Within the NAT region, the area over the ocean and northwards towards the North Pole is designated as airspace in which a minimum standard for air navigation has been specified. This is known as the NAT Minimum Navigation Performance Specification Airspace (MNPSA).

Considerations

The problems are threefold:

Firstly, the North Atlantic suffers from abnormally high levels of traffic density with modern turbine engined aeroplanes wanting to fly at the same flight levels, and an economically generated tidal flow of traffic.

Secondly, the meteorological situation over the Ocean 'standardizes' the polar frontal jet stream and the sub-tropical jet stream at about 55°N and 40°N respectively. The jet stream flow is always from west to east.

Finally, the total absence of ground based navigation aids means that navigation accuracy will not be as good as over land and allowances in separation will need to be made by the ATC authorities.

Transoceanic Navigation Problems

No aerodromes.
No ground based radar.
No fixed radio navigation beacons (VOR; NDB etc.)
Outside of VHF radio range hence reliance on HF communications.

The Airspace

The concentration of the traffic mean that all traffic flying across the North Atlantic is required **to fly IFR**. To this extent, all the airspace is classified as **class A between FL55 and FL660**. The airspace is defined as an Oceanic Control Area (OCA). In practice, there are 5 NAT OCAs encompassing the MNPSA

MNPS Authority

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Operators of aircraft flying within the MNPSA are required to have authority approval (stated on the AOC) requiring the aircraft to be able to navigate in accordance with the relevant RNP.

Approval for MNPS will be indicated to Air Traffic by inserting the letters **SX** in item 10 of the flight plan.

RVSM

MNPS Airspace extends from Flight Level (FL) 285-420, incorporating the RVSM FLs 290-410. Aircraft flying in MNPSA must also be authorized to fly RVSM, and must comply with the altimetry Minimum Aircraft System Performance Specification (MASPS).

Due to the tidal nature of the majority of North Atlantic traffic during the times of the Organized Track System (OTS) **all FL are available in both directions in RVSM airspace**. Outside the OTS standard RVSM FL apply.

The minimum equipment requirement for RVSM flight is as follows;

1. **Two independent barometric altimeters** agreeing to within +/- 200 ft
2. **Autopilot** with height hold capability
3. **Altitude deviation alerting system** (if the a/c deviates by +/-300 ft)
4. **SSR** with altitude alerting mode (mode C)
5. The altimeters must be checked prior to entering the NAT OCA.
6. Pilots must report when reaching any new cruising level, and if a deviation of more than 300 ft occurs it must be reported to ATS, with a subsequent written report post flight.

Navigation System Requirements

Minimum **2 independent Long Range Navigation Systems** (LRNS) to cater for failure of one system. Approved systems may be:

1. GNSS (GPS)
2. INS
3. IRS

Each system must be capable of providing **continuous position, track and speed** information.

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In addition, all turbine powered aeroplanes with MTOM > 5700 kg, are required to carry and operate **ACAS II** in the NAT region.

The requirement for 2 LRNS covers the case of a failure in one system and each system must have a failure warning indication.

NAT Tracks

The most desirable routes are the minimum time tracks (MTT).

To accommodate all flights going the same way at approximately the same time, a set of roughly parallel tracks is established with lateral separation based on the MNPS RNP. These tracks are called organized tracks and the overall concept is called the **organized track system** (OTS).

Communications: Due to the distances involved, the present primary method of communication is **HF SSB voice**. **Long range VHF** is available to aircraft when within 250 NM of land, and **VHF** is also used for the delivery of oceanic clearances to aircraft prior to entering the OCAs.

Position Reports

The system of ATC used across the NAT region is **procedural ATC**.

The routes are organized to cross whole **5° of longitude** at whole degrees of latitude. Likewise, whenever specified, north/ south routes cross whole **10° of latitude** at whole degrees of longitude. This meets the ICAO requirement for position reports to be made at intervals not greater than 1 hour.

Radio Failure in the North Atlantic Area

In the case of radio failure prior to exiting the NAT region, the pilot shall **maintain the last received and acknowledged oceanic clearance**, **including level and speed**, to the last specified oceanic route point, normally landfall, then:

1. **Cleared On Filed Flight Plan Route:** Continue on the filed flight plan route. After passing the last specified oceanic route point; the pilot shall conform to the relevant state procedures/ regulations.

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2. **Cleared Other Than Filed Flight Plan Route:** After passing this point, the pilot shall conform to the relevant state procedures/regulations and rejoin the filed plan route.

Initial Clearance

At least 40 minutes before entering the NAT airspace (above FL55) pilots are required to obtain an oceanic clearance directly from the initial OACC.

Strategic Lateral Offset Procedure (SLOP)

If an aeroplane flying in the NAT MNPSA encounters **wake turbulence** and the pilot considers it necessary, a procedure has been established to allow the aeroplane track to be offset from that of the aeroplane causing the turbulence. The procedure is called the **SLOP** and offers three options.

The **first** allows the aeroplane to be flown along the cleared track **centre line**, the **second** to offset by **1 NM right**, and the **third** to offset by **2 NM to the right** only.

Deviation around Severe Weather

If the aircraft is required to **deviate laterally from track to avoid weather** (e.g. thunderstorms), the pilot should request a **revised clearance** from ATC and obtain essential traffic information prior to deviating.

However, if for any reason such prior revised ATC clearance cannot be obtained, the contingency procedures described below should be adopted:

a) If possible, deviate away from the organized track or route system;

b) Establish communications with and alert nearby aircraft broadcasting, at suitable intervals: aircraft identification, flight level, aircraft position and intentions, on the frequency in use (when VHF) and on frequency 121.5 MHz

c) Watch for conflicting traffic both visually and by reference to ACAS/TCAS

d) Turn on all aircraft exterior lights;

e) For deviations of less than 10 NM, aircraft should remain at the level assigned by ATC;

f) For deviations of greater than 10 NM, when the aircraft is approximately 10 NM from track, initiate a level change of +/-300 ft.

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Unable to Obtain Revised Clearance

If a pilot of an aeroplane is unable to obtain a revised ATC clearance, the aeroplane should leave its assigned route or track by turning **at least 45°** to the right or left whenever this is possible.
When able to maintain assigned flight level:

Turn to acquire and maintain in either direction a track laterally separated by 15 NM from its assigned route or track, and

If above FL410, climb or descend 300 m (1000 ft), or

If below FL410, climb or descend 150 m (500 ft), or

If at FL410, climb 300 m (1000 ft) or descend 150 m (500 ft).

Required Navigation Performance (RNP)

RNP is a numerical representation of the navigational accuracy required within ATS airspace of a State, and is prescribed on the basis of regional air navigation agreements (RANs).

It is based on a 95% containment factor, implying that an aeroplane will be within the required RNP for a period of not less than 95% of the time the aircraft is within the airspace concerned. The state is responsible for specifying the RNP value for its airspace.

*The **RNP factor** relates to navigational accuracy relating to the aircraft plotted position in nautical miles. For instance, **RNP4** implies that the aircraft will be within 4 NM of the plotted position for 95% of the time the aircraft is within the airspace concerned.

The applicable RNP factors are: RNP1; RNP4; RNP10; RNP12.6; RNP20. All ATS airspace is classified for RNP.

*Where **VOR/DME** is used for airways or RNAV navigation, the RNP specified is **RNP5**. Within the classification of RNP however, RNP5 does not exist.

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The use of **radar** permits RNP1 whereas **GPS** theoretically offers a relative value of RNP0.3.

A good example of how RNP is used is the track spacing used for the NAT tracks in the MNPSA of the North Atlantic Oceanic regions. The airspace is classified as RNP20 therefore the aircraft flying the routes will be within 20 NM of the plotted position for not less than 95% of the flight time. This means that the airspace reserved to a NAT track must be 20 NM either side of the specified route. The additional safety 'buffer' will be equal to the RNP so the track spacing will be 20 + 20 + 20 = 60 NM.

Within PBN there are two main categories of navigation methods:

1. area navigation (RNAV) and
2. required navigation performance (RNP).

For an aircraft to meet the requirements of RNAV, a specified RNAV accuracy must be met 95 percent of the flight time.

RNP is an RNAV system that includes onboard performance monitoring and alerting capability (for example, RAIM).

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There are several potential **advantages of RNAV routes and procedures**:
(a) Time and fuel savings;
(b) Reduced dependence on radar vectoring, altitude, and speed assignments allowing a reduction in required ATC radio transmissions; and
(c) More efficient use of airspace.

RNAV Navigation Specifications (Nav Specs)

Nav Specs are **a set of aircraft and aircrew requirements** needed to support a navigation application within a defined airspace concept.

RNAV 1. Typically, RNAV 1 is used for **DPs and STARs** and appears on the charts. Aircraft must maintain a total system error of not more than 1 NM for 95 percent of the total flight time.

RNAV 2. Typically RNAV 2 is used for en-route operations unless otherwise specified. **T-routes and Q-routes** are examples of this Nav Spec. Aircraft must maintain a total system error of not more than 2 NM for 95 percent of the total flight time.

RNAV 10. Typically RNAV 10 is used in oceanic operations.

RNP is RNAV with onboard navigation monitoring and alerting.
RNP is also **a statement of navigation performance necessary for operation** within a defined airspace.

Q: What are RNAV specification with regard to aircraft approach an aerodrome?
Ans. RNAV Route to Initial approach fix RNAV5;
IAF to Final approach FIX RNAV1;
FAF to Landing RNAV 0.3;
Missed approach again is RNAV1

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RNP Level	Typical Application	Primary Route Width (NM) – Centerline to Boundary
0.1 to 1.0	RNP AR Approach Segments	0.1 to 1.0
0.3 to 1.0	RNP Approach Segments	0.3 to 1.0
1	Terminal and En Route	1.0
2	En Route	2.0
4	Projected for oceanic/remote areas where 30 NM horizontal separation is applied.	4.0
10	Oceanic/remote areas where 50 NM lateral separation is applied.	10.0

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RNAV 10 Operations

*RNAV 10 (RNP 10) is required for operations in **oceanic or remote airspace**. The RNAV 10 navigation specification **does not require on-board performance monitoring and alerting**.

Aircraft-based augmentation system (ABAS) - An augmentation system that augments and /or integrates the information obtained from other GNSS elements with information available on board the aircraft.

***Area navigation (RNAV)** - A navigation method that allows aircraft to operate on any desired flight path within the coverage of ground- or space-based navigation aids, or within the limits of the capability of self-contained aids, or a combination of these.

Navigation aid infrastructure

- RNAV 10 (RNP 10) was developed for operations in **oceanic and remote areas** and **does not require any ground-based navigation infrastructure** or assessment.
- The minimum route spacing where RNAV 10 (RNP 10) is utilized is **50 NM**.
- If both aircraft contains CPDLC and ADS-C and /or RNP4 approved, then longitudinal spacing can be 30 NM.

AIRCRAFT REQUIREMENTS:

Navigation systems-The RNAV 10 (RNP 10) navigation specification requires that aircraft operating in **oceanic and remote areas** be equipped with **at least *two independent and serviceable long range navigation systems*** *(LRNS),* comprising an INS, an IRS/FMS, or a GNSS.

Accuracy - the lateral total system error/along-track error must not exceed + 10 NM for at least 95% of the total flight time.

Flight plan designation

The letter **"R"** must be inserted in item 10 of ICAO flight plan to indicate the aircraft and operator have been approved on routes where RNP is a requirement for operation.

Specific capability for RNAV 10 (RNP 10) must be indicated in item 18 of ICAO.

En-route

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- At least two LRNSs capable of satisfying this navigation specification must be operational at the oceanic entry point.
- If this is not the case, then the pilot should consider an alternate route which does not require that particular equipment or having to make a diversion for repairs.

RNAV 5 Operations

Although RNAV 5 specification also **does not include requirements for on-board performance monitoring and alerting** {like RNAV 10}, it **does require that the on-board equipment keeps a lateral and longitudinal navigation accuracy on route of + 5 NM** or better during 95% of the total flight time.

Navigation aid

RNAV 5 navigation is used on **Q routes**.

RNAV 5 operations are **based on the use of RNAV equipment** which automatically determines the aircraft position in the horizontal plane using input from one sensor or a combination of the following types of position sensors:

1) VOR/DME;
2) DME/DME;
3) INS or IRS; and
4) GNSS.

ATS communications and surveillance

Direct pilot to ATC voice communication is required.

RNAV 5 system requirements

Accuracy: requires a track keeping accuracy equal to or better than + 5 NM during the 95% of the flight time.

Contingency procedures: The pilot must **notify ATC** when the RNAV performance ceases to meet the requirements for RNAV 5.

In the event of communications failure, the pilot should **continue with the flight plan in accordance with the published "lost communications" procedure.**

RNP 4 Operations

RNP 4 also used in **oceanic or remote airspace**

Automatic dependent surveillance -contract (ADS-C) - A means that allows the ground system and the aircraft to establish, through data link, the conditions of an ADS-C agreement.

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To implement ICAO Global air navigation plan for CNS/ATM systems, RNP 4 requires lateral and longitudinal separation minima of **30 NM** in RNP 4 airspace.

Navigation aid infrastructure

RNP 4 was developed for operations in **oceanic and remote areas**, therefore, it **does not require any ground-based navaid infrastructure**.

The **GNSS is the primary navigation sensor** that supports RNP 4, either as a stand-alone navigation system or as part of a multi-sensor system.

RNP 4 may be used to support the application of en-route separation/spacing standards of **less than 30 NM in continental airspace**, provided the DGCA has conducted the necessary safety assessments.

Navigation systems requirements

For RNP 4 operations in **oceanic or remote airspace**, the aircraft must have, **at least *two fully operational and independent long-range navigation systems*** *(LRNS)*.

Accuracy - The lateral total system error/ along-track error must not exceed + 4 NM for at least 95% of the total flight time.

Flight plan designation

a. The letter "R" must be inserted in item 10 of ICAO flight plan.

c. CPDLC and ADS-C systems will also be required when the lateral and/or longitudinal separation standard is 30 NM.

Contingency procedures

If an aircraft cannot continue the flight according to the current ATC authorization or cannot maintain RNP 4 accuracy, **it will not enter into or continue operating in RNP 4 airspace.**

RNP 2 Operations

The RNP 2 navigation specification is primarily intended for a diverse set of en-route applications, particularly **in geographic areas with little or no ground NAVAID infrastructure and with limited or no ATS surveillance.**

The RNP 2 is used to support RNP operations in the en-route phase of flight in **oceanic, remote and continental airspace.**

RNP 2 is applicable to area navigation routes defined by **straight segments**; however, RNP 2 can be associated with fixed radius transition (FRT).

Navigation aid

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a. RNP 2 navigation specification requires **GNSS as the primary navigation** sensor, either as a stand-alone navigation system or as part of a multi-sensor system.
b. RNP 2 operations in oceanic and remote airspace require **dual independent long range navigation systems.**

Accuracy - the lateral TSE/ along track error must be within ±2 NM for at least 95 per cent of the total flight time.

Contingency procedures

The pilot must **notify ATC** of any loss of the RNP 2 capability (integrity alerts or loss of navigation). If unable to comply with the requirements of an RNP 2 route for any reason, pilots **must advise ATC as soon as possible.**

Q. What is the accuracy of Q routes?
Ans. Q Routes are based on RNAV5. Aircraft with high navigation performance are
allowed to fly the RNP routes. With higher accuracy, more airplanes can be squeezed on an airway. The "Q" routes allow aircraft to aircraft longitudinal separation of **50NM**.

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RNP 1 Operations

1. The RNP 1 navigation specification provides a means to develop routes **for connectivity between the en-route structure and terminal airspace** with no or limited air traffic service (ATS) surveillance.
2. The RNP 1 is used to support RNP operations on **SIDs, STARs and approaches** (initial and intermediate approach segments) up to the FAF /point.
3. The RNP 1 specification is based upon GNSS.

Communications and ATS surveillance

1. The RNP 1 navigation specification is intended in environments **where ATS surveillance is limited or not available.**
2. RNP 1 SIDs and STARs are primary intended to be conducted in direct controller pilot communication (DCPC) environments.
3. The default alerting functionality switches between terminal alerting (+1 NM) and en-route alerting (+2 NM) **at 30 miles** from the ARP.

Vertical navigation (VNAV)

In VNAV, the system allows the aircraft to fly level and descent point to point in a vertical linear profile path that is kept in an on board navigation database.

Accuracy - total lateral system error/ along-track error must not exceed + 1 NM during at least 95% of total flight time. In order to meet the accuracy requirement, 95% of the flight technical error (FTE) must not exceed 0.5 NM.

Contingency procedures

The pilot must **notify ATC** of any loss of the RNP capability (integrity alerts or loss of navigation), together with the proposed course of action. If unable to comply with the requirements of an RNP 1 SID or STAR for any reason, **pilots must advise ATS as soon as possible.**

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RVSM

RVSM Airspace: RVSM airspace is any designated airspace/route between FL 290 and FL 410 (both levels inclusive) where aircraft are separated vertically by 1000 ft
(300 m).

Indian Airspace and neighboring airspace has become RVSM airspace between FL 290 and FL 410, inclusive of FL 290 and FL 410 on 27thNovember, 2003.

BASIC REQUIREMENTS

1. The operator and the operator's aircraft comply with CAR;
2. The operator is authorised by DGCA to perform RVSM operations.

OPERATIONAL REQUIREMENT:

Each aircraft type group utilised by an operator shall be **capable of Height keeping performance** which does not exceed a mean **Total Vertical Error of 25m** (80ft).

AIRCRAFT SYSTEMS:

1. Two independent altitude measurement systems shall be installed.
2. Each system shall be composed of the following elements:
   - Cross-coupled static source/system, with ice protection.
   - Equipment for providing a digitally encoded signal
   - One SSR transponder with an altitude reporting system.
   - Airborne Collision Avoidance System (ACAS II).
   - An altitude deviation alerting system.
   - An automatic altitude control system.

Flight Envelopes

The RVSM operational flight envelope, is the Mach number W/δ and altitude ranges over which an aircraft can be operated in cruising flights within the RVSM airspace. The RVSM operational flight envelope may be divided into two parts:-

1. Full RVSM Flight Envelope: The full envelope will comprise the entire range of operational Mach number, W/δ and altitude values over which the aircraft can be operated within RVSM airspace.

2. Basic RVSM Flight Planning Envelope: The boundaries for the Basic envelope are the same as those for the full envelope except for the upper Mach boundary.

PROCEDURES PRIOR TO RVSM AIRSPACE ENTRY

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The following equipment shall be operating normally at entry into RVSM airspace:

a. Two primary altitude measurement systems.
b. One automatic altitude-control system.
c. One altitude-alerting device. When changing levels, the a/c shall not be allowed to overshoot or undershoot the cleared flight level by more than 45 m (150 ft).
d. One operating transponder with operational Mode 'C'

***Contingency procedures after entering RVSM airspace are:**

1. The pilot shall notify ATC of contingencies (equipment failures, weather) and co-ordinate a plan of action appropriate to the airspace concerned.

Examples of equipment failures, which shall be notified, to ATC are:

- failure of all automatic altitude-control systems aboard the aircraft;
- loss of redundancy of altimetry systems,
- loss of thrust on an engine necessitating descent; or
- any other equipment failure affecting the ability to maintain cleared flight level (CFL);

2. The pilot should notify ATC when encountering greater than moderate turbulence.
3. If unable to notify ATC and obtain an ATC clearance prior to deviating from the assigned CFL, the pilot shall follow the established contingency procedures and obtain ATC clearance as soon as possible.

Q: If a pilot is unable to maintain RVSM, what is the RT phraseology?
Ans. Unable RVSM due Equipment / Due Turbulence.

What is the difference between MEL and MMEL?
Ans. While the MMEL is for an aircraft type family, the MEL is tailored to the operator's specific aircraft and operating environment and may be dependent upon the route structure, geographic location, and number of airports where spares and maintenance capability are available etc. The MMEL cannot address these individual variables or standard terms such as "as required by regulations". It is for these reasons that a MMEL cannot be approved for use as a MEL. MEL does not cover essential Airworthiness items such as Engine, Landing Gear, Flaps, Wings, Flight Control Surface etc. Categories for different classes are:

A: As per remarks **B**: 3 Days **C**: 10 Days **D**: 120 Days

NAVIGATION GENERAL

ICAO approved model of the earth; World Geodetic System 1984 (**WGS 84**).
Navigation is a fundamental process whereby an aircraft is directed from one position to another.

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Great Circle: A circle on the surface of the earth whose centre and radius are those of the earth itself is called a Great Circle.

Rhumb Line is a regularly curved line on the surface of the Earth which cuts all meridians at the same angle - a line of constant direction.

The Rhumba Line between 2 points will always lie nearer to the Equator than the corresponding Great Circle.

Nautical MILE: A nautical mile is that length of arc of a Great Circle which subtends an angle of one minute at the centre of curvature of the Earth's surface.

VARIATION: Variation is the angular difference between the directions of True and Magnetic North at any point.

ISOGONAL: A line on the surface of the Earth joining points of equal magnetic variation is called an Isogonal.

Agonic Line is the line connecting points of zero variation.

Deviation is defined as the angle measured at a point between the direction indicated by a compass needle and the direction of Magnetic North.

Heading is defined as the direction in which the fore and aft axis of the aircraft is pointing; it may be measured from True, Magnetic, or Compass North.

Track is the direction of the aircraft's path over the ground. It may be measured from True or Magnetic North.

Drift is the difference between heading and track.

1:60 Rule?

The 1-in-60 rules is a basic rule of thumb used in air navigation. It states:

One degree offset equals one nautical mile when sixty nautical miles from the origin

The track you intended to fly is called the "**planned track**".

The track you actually make over the ground is called the "**track made good**".

The difference between them is called "**Track Error Angle**".

Drift is the angle between **Heading** and **Track** (either planned track or TMG).

The difference between the planned track and the TMG is the **track error angle**.

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The difference between the heading and the planned track is the expected drift.
The difference between the heading and the TMG is the actual drift experienced.

There are 3 basic techniques for getting back to track:

1. Double Track Angle Error
2. Track Error Angle and Closing Angle
3. Combined Track Error Angle and Closing Angle Single Calculation.

APPLICATIONS OF THE 1 in 60 RULE

The other main applications which may be examined are:

1. Calculation of height on a glide slope.
2. Calculation of rate of descent to maintain a glide slope.
3. Calculation of shallow slopes, particularly runway slopes.
4. Calculation of distance off required track using radio aids such as VOR/DME.

To make good a glide slope of Z°, your rate of descent should be (100 x Z) ft per nautical mile:

3° glide slope 300 feet/Nm

ROD (in feet/min) **= 5 x Ground speed** (in knots) **-** for a **3° glideslope** only
If the glideslope is some angle other than 3°, the easiest way is to solve for a 3° glideslope and then factor the answer by actual glideslope ÷ 3°.
Change in ROD = 5 x change in speed (3° glideslopes only)

SCALE

Scale is the relationship between the length of a line drawn between two positions on a chart and the distance on the earth between the same points.

Three methods of expressing scale are common:

1. **Statement in words**.
2. **Graduated Scale Line**
3. **Representative fraction**

Q: What is Convergency and Conversion Angle? Does convergency increase or decrease towards pole?

Ans. Convergency is defined as the angle of inclination between two selected meridians measured at a given latitude.

All the meridians on the surface of the earth from equator to the pole converge and they make angle at the pole.
At equator meridians are parallels to each other, therefore convergency is zero.

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At the pole they make maximum angle, the angle is change of longitude.

Conversion Angle is the difference between Great Circle direction and Rhumb Line direction joining two given points.

Departure is the distance between two meridians along a specified parallel of latitude, usually in nautical miles.

Q. Why GC track is not a constant track?
Ans. Due earth's convergency.

CHARTS

Charts produced directly from a projection are called perspective or geometric projections.

TYPES OF PROJECTION

There are 3 general types of projection surfaces:

1. **Azimuthal/Plane:** An azimuthal (or 'plane') projection is produced by placing a flat sheet of paper against a point on the earth**.** A common use is to provide charts of the North/South polar regions.
2. **Cylindrical:** Cylindrical projections uses cylinder of paper wrapped around the "reduced earth" and touching the RE at the Equator. Other charts that use cylindrical projections are the Transverse Mercator and the Oblique Mercator.
3. **Conical:** Conical projections involve placing a cone of paper over the Reduced Earth and projecting the graticule onto the cone. Subsequently the cone is slit along one side and the cone can then be opened to produce a flat sheet of paper.

ORTHOMORPHISM / CONFORMALITY

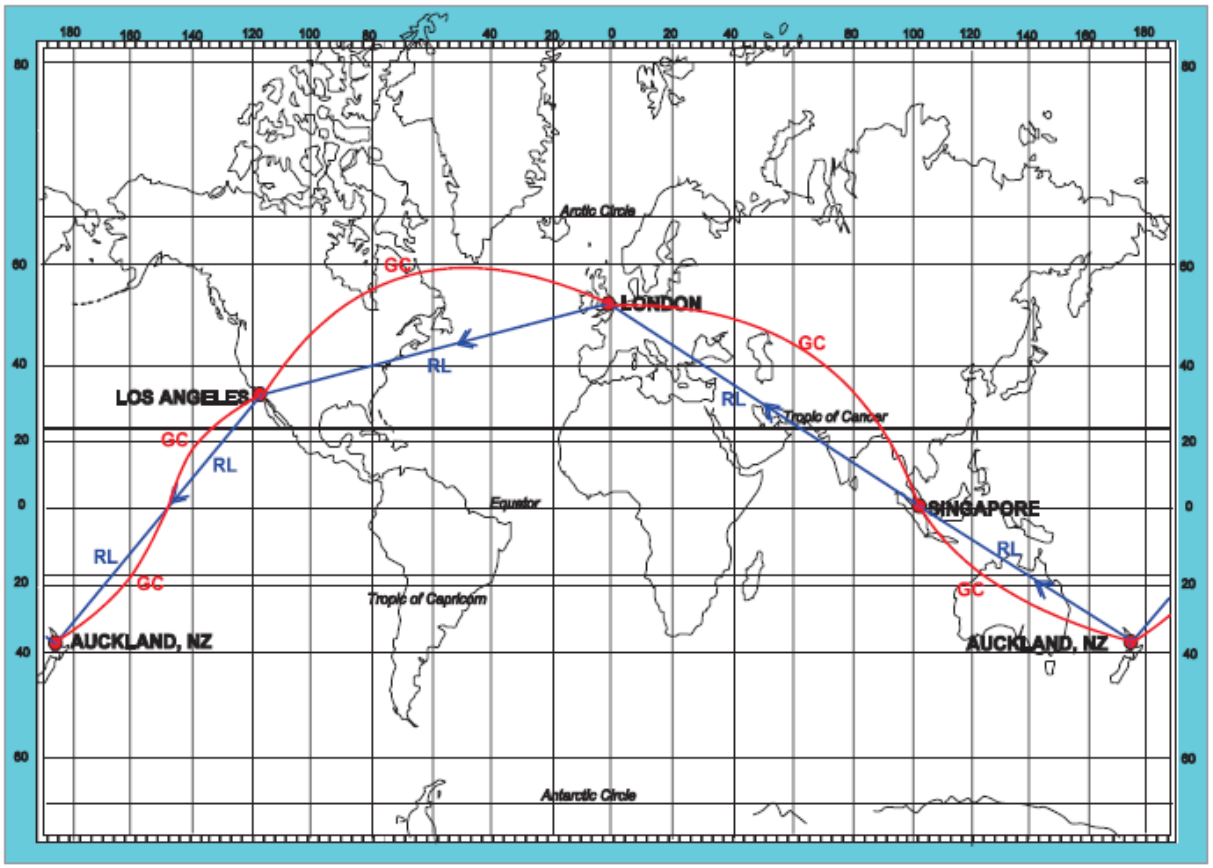
There are two fundamental conditions which must be met to achieve orthomorphism/ conformality.

1: The meridians and parallels on the chart must intersect at right angles.
Condition 2: At any point on a chart, scale should be the same in all directions, or should change at the same rate in all directions.

1. MERCATOR'S CHART

A cylinder of paper is wrapped around the RE, touching the RE at the Equator. Using a light source at the centre of the RE, the graticule is projected onto the cylinder. The cylinder is then 'developed' or opened up to a flat sheet of paper.

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Advantage

1. **The meridians are equally spaced parallel lines**.
2. **The straight line on the chart would be a rhumb line**.

Mercator chart is a **non-perspective** chart.

USES OF A MERCATOR CHART

1. Plotting charts of areas up to 75° latitude.
2. Topographical charts of equatorial areas, where scale distortion is small.
3. Meteorological synoptic charts of equatorial charts.

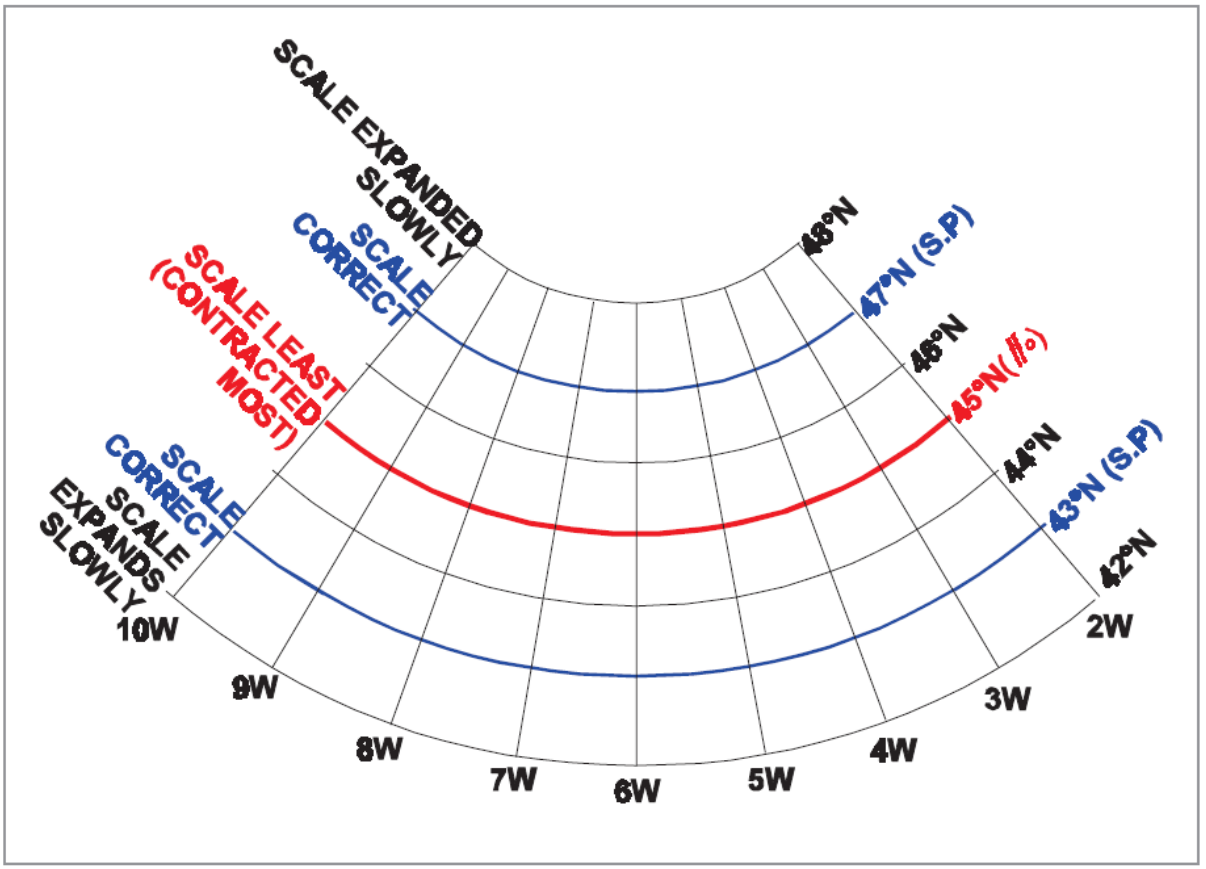
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2. LAMBERT'S CONFORMAL CONICAL CHART

WHY CONICAL CHARTS?

Mercator chart has 2 limitations.

1. Great Circles are not projected as straight lines.
2. The chart is not constant scale.



If the scale were to remain constant on a chart, we could measure distances with a ruler instead of having to use a pair of dividers.

Projection:

A cone is placed over a reduced earth, in such a way that the cone is tangential with the reduced earth along a parallel of latitude. The apex of the cone will lie on the extended line of the earth's axis.

Light source is kept at the centre of the reduced earth.

ADVANTAGES

1. Great Circles can be treated as straight lines for all practical purposes.
2. Radio bearings are Great Circles, so can be plotted without application of conversion angle.
3. If scale error is small, a graduated ruler can be used to measure distances.

PLOTTING ON A LAMBERT CHART

Plotting bearings on a Lambert chart is less complicated than on a Mercator because Great Circles (radio waves) are straight or near-straight lines, and so there is no conversion angle to apply.

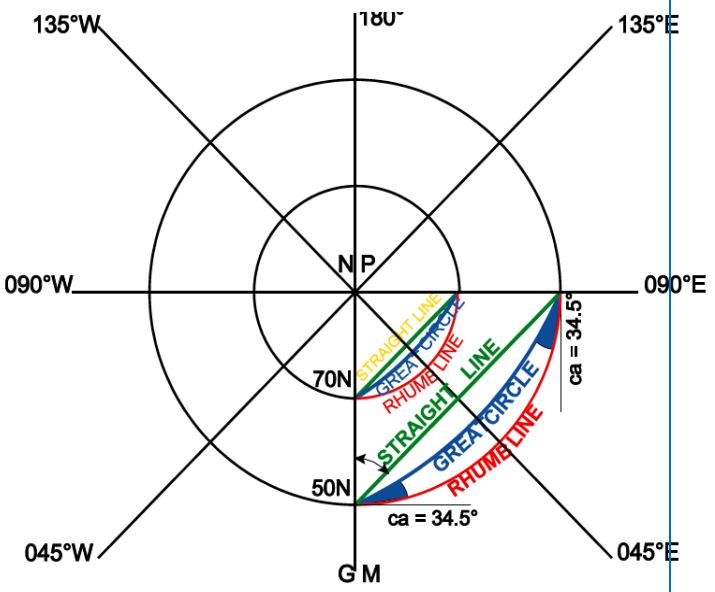
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3. POLAR STEREOGRAPHIC PROJECTION

The **polar stereographic projection** is, of those discussed, the only geometric projection, i.e. true **perspective** projection.

A flat surface is used, touching one Pole. The light source is positioned at the other Pole.

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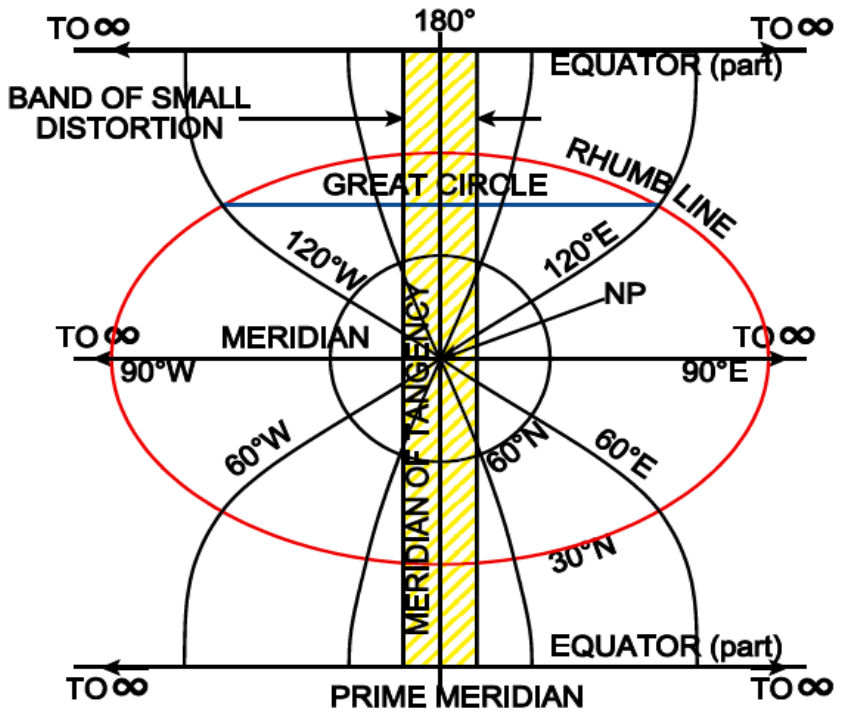
USES OF A POLAR STEREOGRAPHIC CHART

The chart is used mainly for plotting in Polar regions.

	Mercator's Chart	Lambert's Conformal Conical	Polar Stereographic
Light Source	At centre	At centre	At opposite pole
Scale	Correct @equator Changes @Sec of Lat Considered constant up to 8 degree from equator	Correct @Standard Parallel Expands outside Consider constant up to 16 deg change bet std parallels.	Correct @pole **Scale expands @secant squared of half the co-lat.** Contant upto 78 deg.
Orthomorphic	Yes. All charts used for navigation must be	Yes	Yes
Rhumbh line	Straight line	Concave to pole except meridians	Concave to the pole
Great Circle	Concave to equator except meridian & equator	Curve concave to Parallel of origin	Concave to the pole with less curvature. Considered straight line upto 70 deg.
Parallel of origin	Equator	Halfway between the Standard Parallels	Pole/ 90 degree
Chart Convergence	Zero	Constant all over chart	Correct only @pole Constant all over chart

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4. TRANSVERSE MERCATOR'S CHART



PROPERTIES OF A TRANSVERSE MERCATOR CHART	
Scale	Correct Along Datum Meridian. Elsewhere expands as sec (Great Circle Distance E / W)
Orthomorphic	Yes. All charts used for navigation must be
Graticule	Datum meridian, meridian perpendicular to it and the Equator are straight lines. Other meridians are complex curves. Parallels are ellipses (Equator is straight lines - limiting case of ellipse)
Shapes	Distorted away from Datum Meridian
Chart Convergence	Correct at the Pole and Equator
Rhumb lines	Complex curves except Datum Meridian, meridian perpendicular to it and the Equator
Great Circles	Complex curves except Datum Meridian, meridian perpendicular to it, Equator and Great Circles perpendicular to the Datum Meridian

CRITICAL POINT

The **Critical Point (CP) is the decision point between two airfields from which it would take the same time to fly to either airfield.**

In case of a cabin fire or passenger with heart attack you will have to decide whether it is quicker to return to your departure point or to continue to your destination.

1. CP is independent of fuel endurance.
2. In still air, CP is midway between departure point and destination.
3. CP always moves into wind from the still air midpoint.
4. If the wind reduces, the CP will move towards the midpoint.
5. The effect of increasing TAS is the same as that of reducing wind.

The **POINT OF SAFE RETURN (PSR)**, or **POINT OF NO RETURN (PNR)**, is the furthest point along a planned route to which an aircraft can fly and return to the departure airfield, or departure alternate, within the SAFE ENDURANCE of the aircraft.

If the weather at your destination deteriorates below landing minima, and you have no viable weather diversion.

Some simple PNR rules are:

1. Maximum distance to PNR is achieved in still air conditions.
2. If a wind component increases, the distance to PNR decreases.
3. An increase in TAS has the same effect as a reduction in wind component - an increase in the distance to PNR.

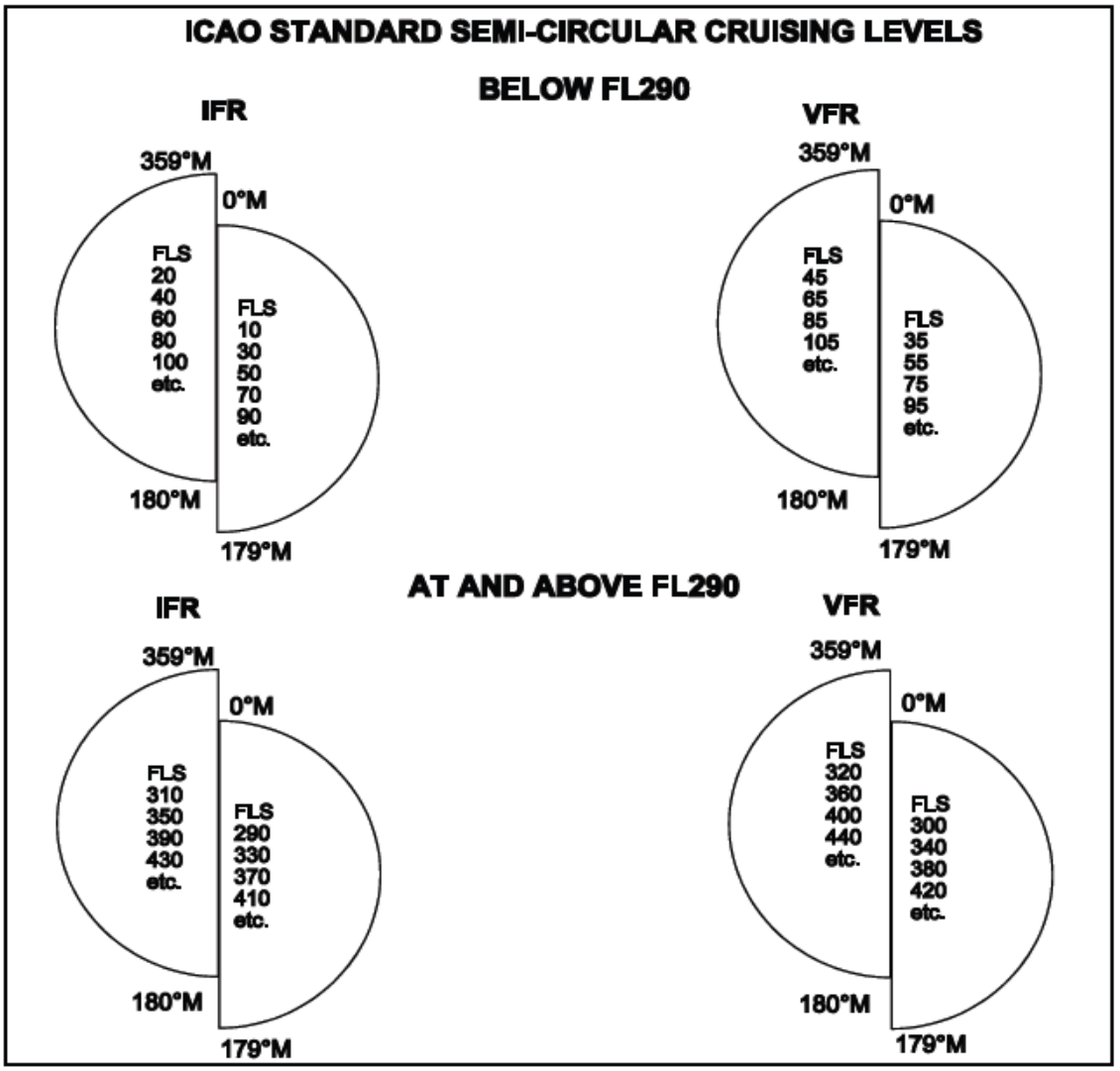
***NAUTICAL AIR MILES:**

NAM is a measure of the air distance flown by an aircraft. i.e. the distance flown at the True Air Speed (TAS).

In still air the NAM flown are equal to the Nautical Ground Distance (NGM) flown.

NGM is the distance flown by the aircraft over the ground, as may be measured on a chart.

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The choice of safe Flight Level is determined by the aircraft's planned **Magnetic track** (not heading).

AERODROME ELEVATION: The elevation of the highest point of the landing area.

ALTITUDE: The vertical distance of a level, point or an object considered as a point, measured from mean sea level (QNH).

DECISION ALTITUDE (DA) or DECISION HEIGHT (DH):

A specified altitude or height in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

DA is referenced to means sea level; **DH** is referenced to the threshold elevation.

FLIGHT PLANNING

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FLIGHT LEVEL (FL): A surface of constant atmospheric pressure which is related to a specific pressure datum 1013.2mb, and is separated from other such surfaces by specific pressure levels.

HEIGHT: The vertical distance of a level, a point or an object considered as a point, measured from a specified datum(QFE).

MINIMUM DESCENT ALTITUDE/HEIGHT: A specified altitude/height in a **non-precision approach or circling approach** below which descent may not be made without visual reference.

THRESHOLD: The beginning of that portion of the runway usable for landing

TRANSITION ALTITUDE: The altitude at or below which the vertical position of the aircraft is controlled by reference to altitudes.

Minimum TA in india is 4000 feet.

TRANSITION LAYER: The airspace between the transition altitude and the transition level.

TRANSITION LEVEL: The lowest flight level available for use above the transition altitude.

QFE: The atmospheric pressure at aerodrome elevation, or at runway threshold.

QNH: The altimeter sub-scale reading to obtain elevation when on the ground.

QNE is the height indicated on landing at an aerodrome when the altimeter sub-scale is set to 1013.2hPa.

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MINIMUM EN-ROUTE IFR ALTITUDE (MEA)

The lowest published altitude between radio fixes that meets obstacle clearance requirements between those fixes and assures acceptable navigational signal coverage.

The MEA applies to the entire width of the airway, segment or route between the radio fixes defining the airway, segment or route.

MINIMUM OBSTRUCTION CLEARANCE ALTITUDE (MOCA):

The lowest published altitude in effect between radio fixes on VOR airways which meets obstacle clearance requirements for the entire route segment and assures acceptable navigational signal coverage only within 22nm of a VOR.

MINIMUM OFF-ROUTE ALTITUDE (MORA):

This is an altitude derived by Jeppesen. The MORA provides known obstacle clearance within 10nm of the route centre-line.

MAXIMUM AUTHORISED ALTITUDE (MAA):

A published altitude representing the maximum usable altitude or flight level for an airspace structure or route segment.

GRID MINIMUM OFF-ROUTE ALTITUDE (Grid MORA) — An altitude derived by Jeppesen or provided by State Authorities. The Grid MORA altitude **provides terrain and man-made structure clearance within the section outlined by latitude and longitude lines**. MORA does not provide for navaid signal coverage or communication coverage.

The Grid MORA values clear all terrain and manmade structures by 1000ft in areas where the highest elevations are 5000ft amsl or lower.

MORA values clear all terrain and manmade structures by 2000ft in areas where the highest elevations are 5001ft amsl or higher.

When a Grid MORA is shown as **"Unsurveyed"** it is due to incomplete or insufficient information.

Grid MORA values followed by **+/-** denote doubtful accuracy, but are believed to provide sufficient reference point clearance.

WAKE TURBULENCE CATEGORY

H – HEAVY (136,000 kg or more)
M – MEDIUM (less than 136,000 kg but more than 7000 kg)
L – LIGHT (7000 kg or less based on certified T/o weight)*

FLIGHT RULES

I if IFR
V if VFR
Y if IFR first) and specify in Item 15 the point
Z if VFR first) or points where a change of flight Rules is planned.

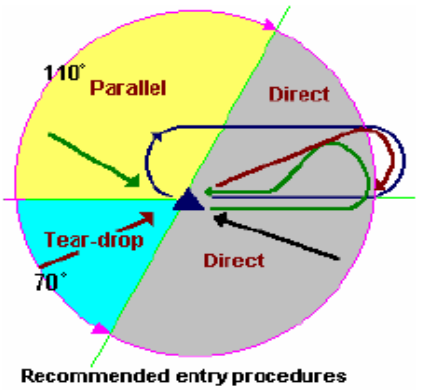
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FUEL PLANNING

PISTON PROP AIRCRAFT	TURBOJET AIRCRAFT
WHEN ALTERNATE RUNWAY IS REQUIRED (IFR FLIGHT):	
FUEL TO DESTINATION + FUEL TO ALTN + 45 MIN OF HOLDING FUEL	FUEL TO DESTINATION + FUEL TO ALTN + 30 MIN OF HOLDING FUEL OVER ALTERNATE AERODROME @ 1500 FT AGL + 5% OF TRIP AS CONTINGENCY FUEL
WHEN ALTERNATE IS NOT REQUIRED (VFR FLIGHT)	
FUEL TO DSTN + 45 MIN OF HOLDING FUEL	FUEL TO DSTN + 30 MIN OF HOLDING FUEL @1500 FT AGL
WHEN ALTN IS NOT AVAILABLE (ISOLATED AERODROME) FUEL TO DSTN + 45 MIN OF FUEL AND 15% OF TRIP FUEL OR 2 HOURS OF FUEL AT NORMAL CONSUMPTION (WHICHEVER IS LESS)	WHEN ALTN IS NOT AVAILABLE (ISOLATED AERODROME) FUEL TO DSTN + 2 HOURS OF FUEL AT NORMAL CONSUMPTION

Holding Patterns and Procedures

A holding pattern provides a protected airspace for a safe operation during the hold. These are the **direct, parallel and tear-drop** entries.



Each holding pattern has a fix, a direction to hold FROM the fix, and an airway, bearing, course, radial, or route on which the aircraft is to hold.

Standard holding pattern - all turns made to right.

The inbound leg of a holding pattern is 1 minute at & below 14,000 ft. and 1.5 minutes above 14,000 ft.

Holding Summary

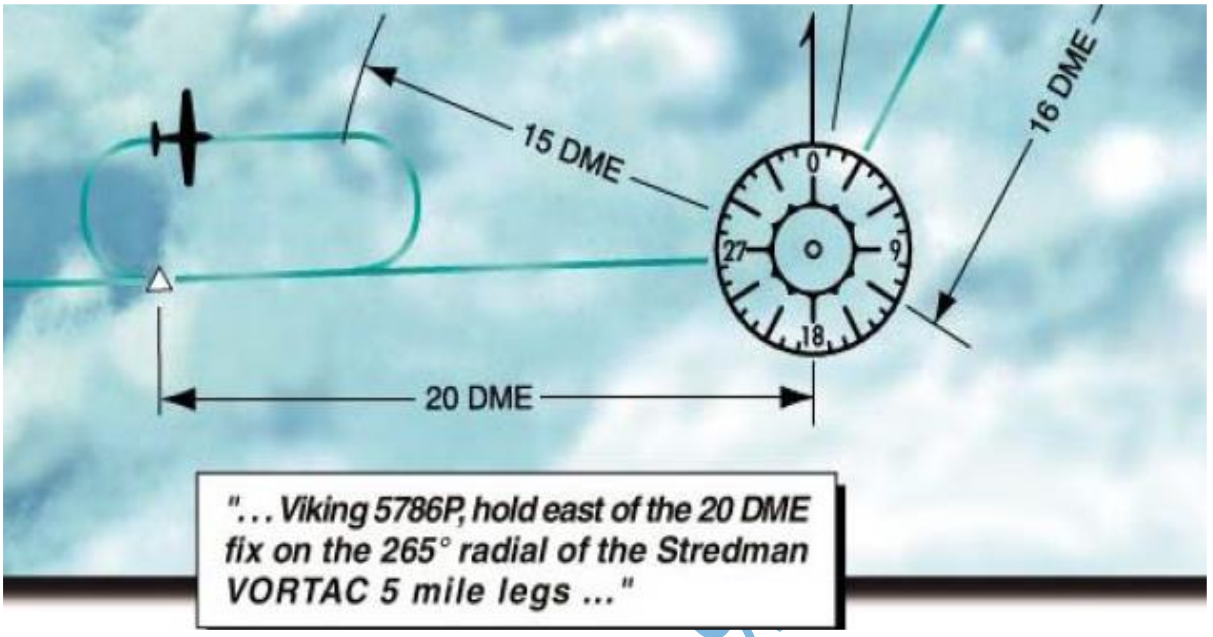
• A holding pattern is a time delay used by ATC to help maintain separation and smooth out the traffic flow.

• Holding pattern size is directly proportional to aircraft speed; doubling your speed doubles the size of the holding pattern.

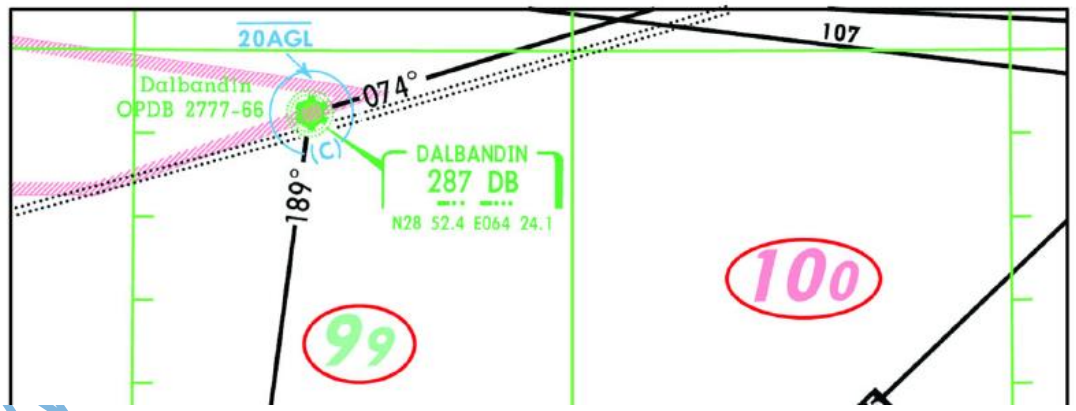
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• Turns are to the right in standard holding patterns, and to the left in nonstandard holding patterns.

For patterns using DME, the clearance gives the outbound leg length in nautical miles.



The colors for depicting GRID MORA values will be adjusted to:
- Values below 10,000 ft will be depicted in green color;
- Values 10,000 ft and above will be depicted in magenta color.



JEPPESEN EN-ROUTE CHARTS

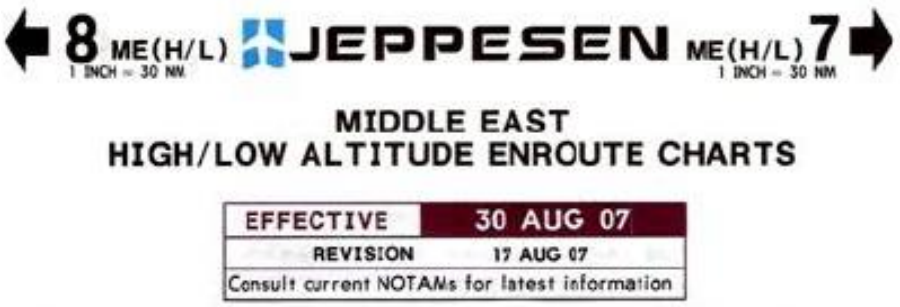
Most of the Jeppesen en-route Charts use the Lambert Conformal Conic projection.

These charts are basically for IFR flights. However Jeppesen has introduced VFR GPS charts using 1:500,000 topographical maps.

Chart Usage:

a. Charts are identified by code letters for world areas covered by a particular series, by parenthetical letters for the altitude coverage and by numbers for the individual chart. For example,

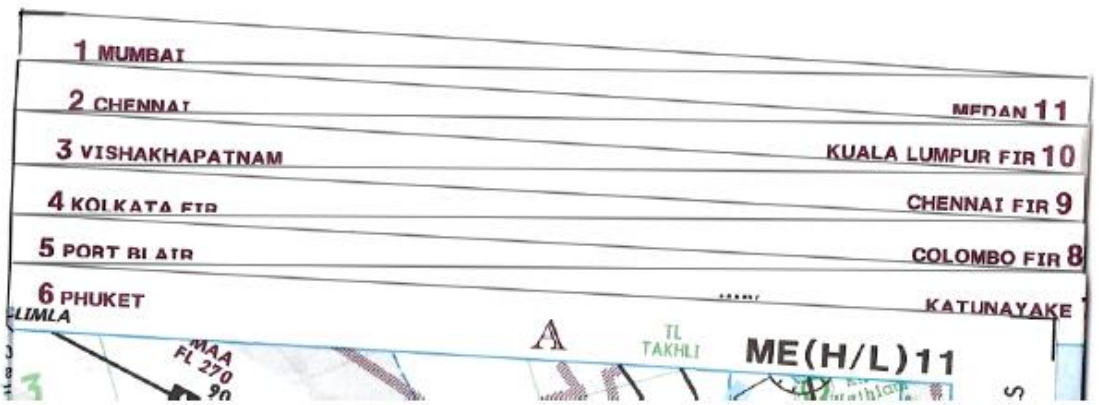
ME(H/L)11, is a chart of the Middle East series covering both high and low altitude operations and is number **11** of the series.



FE(H/L) 2, is a chart of the Far East series covering both high and low altitude operations and is number **2** of the series.

b. The front page of the chart is called the cover panel. To use the low and high altitude charts, we have to first use the small index map on the cover panel.

The low and high/low altitude Jeppesen charts have a unique feature called the **Zigdex**. This zigdex is found on the top of each chart panel on which the major city names are printed.



c. Unless other wise indicated, all bearings and radials are magnetic.
d. En-route distances are in nautical miles.
e. Vertical measurements of elevation are in feet above mean sea level.
f. En-route altitudes are either in feet above mean sea level (based on QNH) or clearly expressed as Flight Levels, based on QNE.
g. All times are Coordinated Universal Time (UTC) unless labeled Local Time (LT).
h. Scale is 1 inch = 30 nm
i. Projection used is Lamberts Conformal Conic.

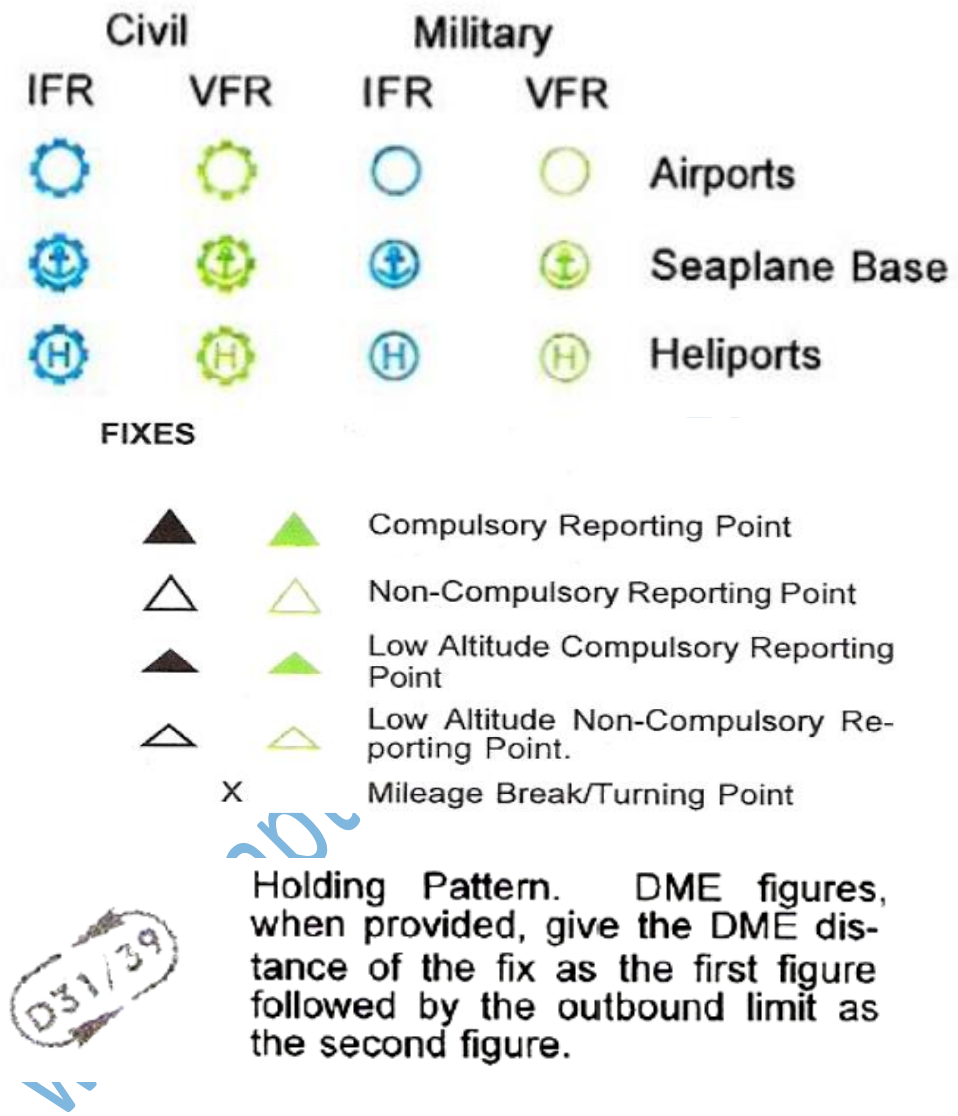
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j. Chart effective dates are printed on the cover panel.

Restricted Airspace:

Restricted airspace areas are enclosed in red slanting lines. They are designated by the Country code, followed by the FIR code, followed by the alphabets 'D', 'P', 'R' in parenthesis.

(a) D – Danger area
(b) P – Prohibited area
(c) R – Restricted area



MDA) OR MDH — A specified altitude or height in a non-precision approach or circling approach below which descent must not be made without the required visual reference.

Segments of approach?

An instrument approach procedure may contain up to five separate segments, which depict course, distance, and minimum altitude. These segments are:

Feeder routes: A route for aircraft to proceed from the en route structure to the IAF, which includes the course and bearing to be flown, the distance, and the minimum altitude.

Initial approach segment: This segment provides a method for aligning the aircraft with the intermediate or final approach segment and to permit descent during the alignment. It begins at an IAF and ends at the intermediate approach segment or intermediate fix (IF).

Intermediate approach segment: This segment positions the aircraft for the final descent to the airport. It begins at the IF and ends at the final approach segment.

Final approach segment:

For a PA or APV, this segment starts where the glideslope intercepts the glideslope intercept altitude plane.

For an NPA, this segment starts at the FAF, final approach point (FAP), or where the aircraft is established on the final approach course. This segment ends at either the designated missed approach point (MAP) or upon landing.

Missed approach segment: This segment starts at the MAP and ends at a point or fix where the initial or en route segment begins.

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LOADING AND CENTRE OF GRAVITY

Centre of Gravity limits:

The CG is not a fixed point it has a range of movement between a maximum forward position and a maximum rearward position which are set by the aircraft manufacturer and cannot be exceeded.

Datum

A point along the longitudinal axis (Centre line) of the aeroplane (or it extension) designated by the manufacturer as the zero or reference point from which all balance arms begin.

Balance Arm

The distance from the aircraft's Datum to the CG position or centroid of a body of mass.

Traffic Load: Originally known as the 'payload', the traffic load is the revenue generating load that pays the salaries and hopefully produces a profit for the operator.

BEM or Tare Weight:

The mass of the aircraft with all its basic equipment plus a declared quantity of unusable fuel and oil.

DOM: the total mass of the aeroplane ready for a specified type of operation, excluding all fuel and traffic load. This mass includes items such as:

1. Crew and crew baggage
2. Catering and removable passenger service equipment
3. Potable water and lavatory chemicals.

MZFM: the max permissible mass of an aeroplane with no usable fuel.

- **Usable fuel includes both Air and Ground fuel.**
- **Ramp Mass = DOM + Usable Fuel + Traffic Load**
- **Ramp Mass = OM + Traffic Load**
- **Takeoff mass = OM +TL – Start and Taxi fuel**
- **Max Traffic Load = MSTOM-DOM-Take-off fuel**
- **RATOM or MATOM = lesser of the MSTOM or PLTOM**
- **RLM or MALM = lesser of the MLM or PLLM**

To calculate the payload:

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RTOW

MLW + Flight Fuel
MZFW + FOB
} Minimum value

Lowest of the three = APS +FOB + Payload

CALCULATION OF CENTRE OF GRAVITY:

Total moment = $(M_1 \times d_1) + (M_2 \times d_2) + (M_3 \times d_3) = M \times d$

$$\textbf{CG position} = \frac{\textbf{Total Moment}}{\textbf{Total Mass}}$$

Nose up moment are positive.

Nose down moment are negative.

Mass is the amount of matter in a body in kilograms and **weight** is the force that the matter exerts on the earth's surface, in Newtons.

Fuel load definitions:

1. **Start and Taxi Fuel**: The mass of fuel used in starting and operating the APU and the main engines and in taxying to the runway threshold for take-off.
2. **Trip Fuel:** This is the mass of the fuel to complete the **T/o run, the climb, the cruise, the descent, the expected arrival procedures, the approach and landing** at the designated airport.
3. **Contingency Fuel:** Fuel carried in addition to the trip fuel for unforeseen eventualities such as avoiding bad weather or having an extended duration at the destination airport. This is usually given as a % of the trip fuel. Contingency fuel is part of the landing mass if not used during the trip.
4. **Alternate Fuel:** The mass of fuel required to carry out a missed approach at the destination airfield, the subsequent climb out, transit to, expected arrival procedures, approach, descent and landing at an alternate airfield.
5. **Final Reserve Fuel:** The minimum fuel that should be in the tanks on landing. Essentially it is a final reserved for unplanned eventualities and should allow a PE a/c to fly for a further 45 minutes or a JE aeroplane to fly for a further 30 minutes*(Air India) at a given height and holding speed.
6. **Additional Fuel:** Only required if the trip, contingency, alternate and final reserve fuels are insufficient to cover the requirements.

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PERFORMANCE

Aerodrome Elevation The elevation of the highest point of the landing area.

Aerodrome Reference Point The aerodrome reference point is the geographical location of the aerodrome and the centre of its traffic zone where an ATZ is established.

Aircraft Classification Number (ACN): This is a value assigned to an aeroplane to show its load force. The aircraft classification number must be compared to the pavement classification number (PCN) of an aerodrome.

Alternate Airport: An airport at which an aircraft may land if a landing at the intended airport becomes inadvisable.
Balanced Field A runway for which the ASDA is equal to the TODA is considered to have a balanced field length.

Clearway An area beyond the runway, not less than 152 m (500 ft) wide, centrally located about the extended centre line of the runway, and under the control of the airport authorities. The clearway is expressed in terms of a clearway plane, extending from the end of the runway with an upward slope not exceeding 1·25%, above which no object or terrain protrudes.

Critical Engine The engine whose failure would most adversely affect the performance or handling qualities of an aircraft.

Damp Runway A runway is considered damp when the surface is not dry, but when the moisture on it does not give it a shiny appearance.

Decision Speed: The **maximum speed** in the take-off at which the pilot can take the first action (e.g. apply brakes, reduce thrust, deploy speed brakes) to stop the aeroplane within the accelerate-stop distance.

It also means the **minimum speed** in the take-off, following a failure of the critical engine at VEF at which the pilot can continue the take-off and achieve the required height above the take-off surface within the take-off distance.

En Route The en route phase extends from 1500 ft above the take-off surface level to 1000 ft above the landing aerodrome surface level for Class B aeroplanes or to 1500 ft above the landing aerodrome surface level for Class A aeroplanes.

Final En Route Climb Speed The speed of the aeroplane in **segment four** of the take-off flight path with one engine inoperative.

Final Segment Speed The speed of the aeroplane in **segment four** of the take-off flight path with one engine inoperative.

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Final Take-off Speed The speed of the aeroplane that exists at the end of the
Gross Performance The average performance that a fleet of aeroplanes should achieve if satisfactorily maintained and flown in accordance with the techniques described in the manual.

Ground Minimum Control Speed V_{MCG}: The minimum speed at which the aeroplane can be demonstrated to be controlled on the ground **using only the primary flight controls** when the most critical engine is suddenly made inoperative and the remaining engines are at take-off thrust. Throttling an opposite engine is not allowed in this demonstration. Forward pressure from the elevators is allowed to hold the nose wheel on the runway, however, **nose wheel steering is not allowed**.

Hydroplaning Speed The speed at which the wheel is held off the runway by a depth of water and directional control through the wheel is impossible.

Landing Gear Extended Speed V_{LE} The maximum speed at which an aircraft can be safely flown with the landing gear extended.

Landing Gear Operating Speed V_{LO} The maximum speed at which the landing gear can be safely extended or retracted.

Net Performance Net performance is the **gross performance diminished** to allow for various contingencies that cannot be accounted for operationally e.g. variations in piloting technique, temporary below average performance, etc.

Pressure Altitude The altitude of an aircraft above the pressure level of 1013.25 hPa. This is achieved by setting the altimeter subscale to 1013 hPa and reading the altitude indicated.

Reference Landing Speed V_{REF} The speed of the aeroplane, in a specified landing configuration, at the point where it descends through the landing screen height in the determination of the landing distance for manual landings.

Runway Threshold The beginning of that portion of the runway usable for landing.

Service Ceiling The pressure altitude at which the rate of climb is reduced to a specified minimum value (approximately 300 ft/min).
Stopway An area beyond the take-off runway, **no less wide than the runway** and centred upon the extended centre line of the runway, **able to support the aeroplane** during an aborted take-off, without causing structural damage to the aeroplane.

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Take-off Distance Available. It is equal to TORA plus any clearway and cannot be more than one and one half times the TORA, whichever is the less.

Take-off Mass The mass of an aeroplane, including everything and everyone contained within it, at the start of the take-off run.

Take-off Safety Speed V_2 A referenced airspeed obtained after lift-off at which the required one

Measured Performance

This is the performance achieved by the manufacturer under test conditions for certification. It utilizes new aeroplanes and test pilots and is therefore unrepresentative of the performance that will be achieved by an average fleet of aeroplanes.

Accelerate-Stop Distance Available means T/o Run Available + Stopway.

Balanced Field means that TODA = ASDA

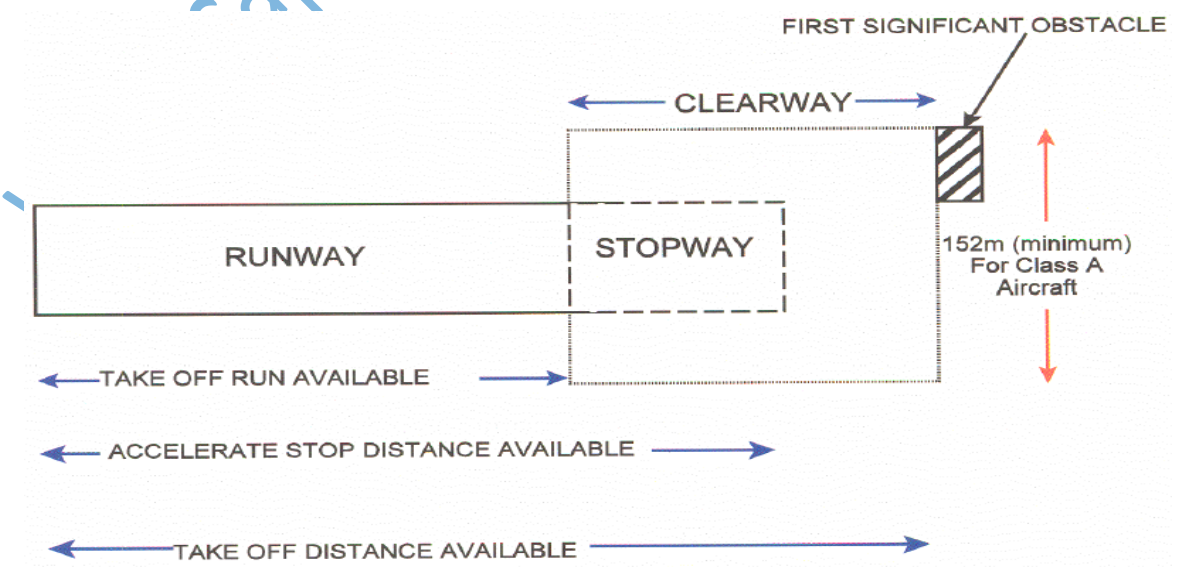
Absolute ceiling means the pressure altitude at which the rate of climb is zero.

Service ceiling means the pressure altitude at which the rate of climb is a defined value.

Climb Gradient means the ratio, in the same units, and expressed as a percentage, of $\frac{\text{Change in Height}}{\text{horizontal Distance Travelled}}$

The **take off part** of the flight is the distance from the brake release point to the point at which the aircraft reaches a 'screen' of defined height.

AVAILABLE DISTANCES:

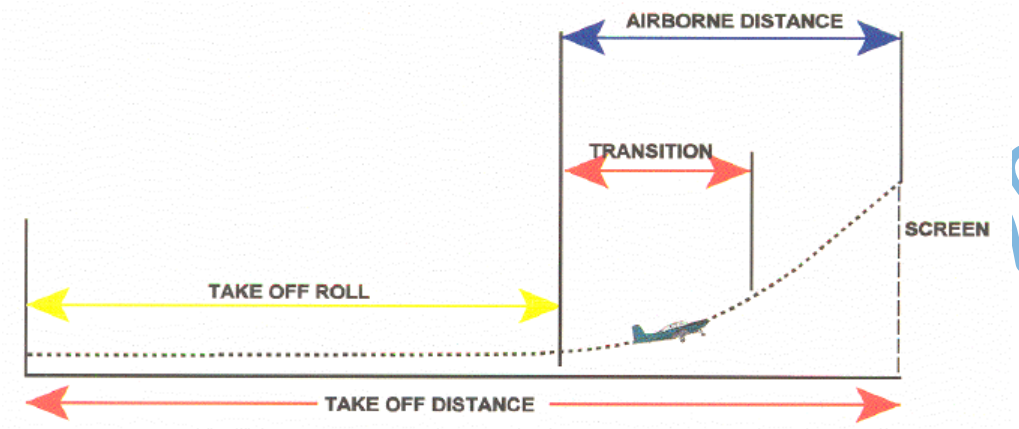


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The **purpose of the clearway** is to ensure that the a/c will not hit any obstacles after it leaves the runway.

REQUIRED DISTANCES for TAKE OFF:

a. The take off roll or ground run
b. The airborne distance to a "screen" of defined height.



Flat Rated Engines:

Flat rated engines are restricted to a maximum thrust even though the engine is capable of producing higher trust.

At temperatures below the flat rating cut off, (typically about ISA +15^{0}C) engine thrust is not affected by temperature.

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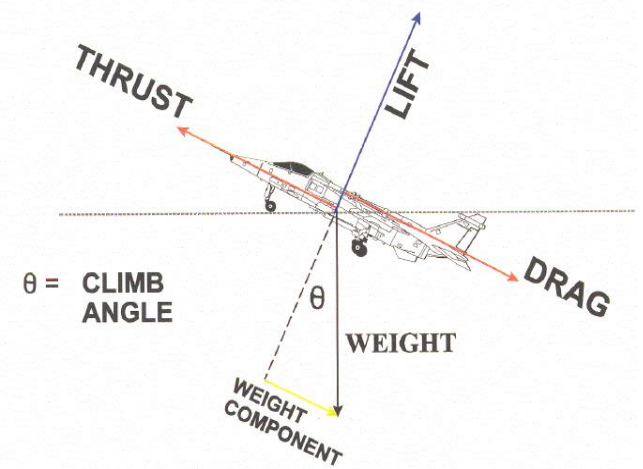
CLIMB

The climb section of aircraft performance deals with the analysis of that stage of flight <u>from the end of the take-off phase to the beginning of the en-route phase.</u>

In a steady climb, the weight has a component along the flight path, adds to the drag force

$T = D + W \sin \Theta$

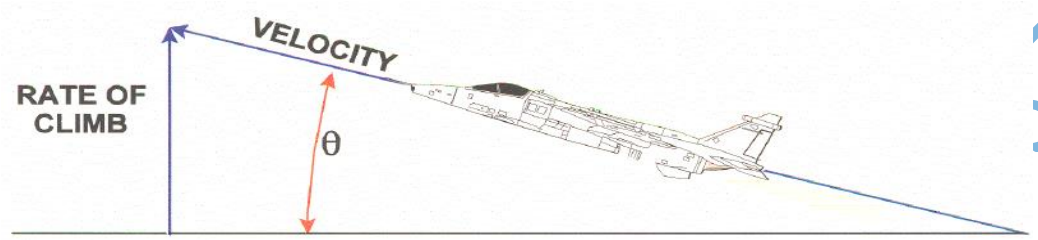
Hence <u>$\sin \Theta = T - D / W$</u>



Climb gradient is the ratio of height gained to distance travelled, and is therefore the tangent of the climb angle.

RATE OF CLIMB:

The rate of climb is the vertical component of the aircraft's velocity and depends on the aircraft's velocity and the climb angle.



Rate of climb = Power available - Power required / W

For a given mass the rate of climb is a maximum when the excess power is greatest. The speed at which this occurs is called $\mathbf{V_y}$.

Ceiling: The gradient of climb and RoC decrease with altitude, and eventually at some altitude become zero. This altitude is called the absolute ceiling.

The absolute ceiling decreases with increasing aeroplane mass, and increasing temperature.

Service ceiling is the altitude at which a specified rate of climb occurs.

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LANDING

LANDING DISTANCE

The **landing stage** of flight is defined as being that stage of flight commencing from 50ft above the landing threshold and terminating when the aeroplane comes to a complete stop.

The 50 ft point is sometimes referred to as the **landing screen height**.

The landing screen height is fixed at 50 ft for all classes of aeroplane unlike the take-off screen height which is 35 ft for class A aeroplanes and 50 ft for class B aeroplanes.

From the approach down to the landing screen height the aeroplane must have attained the **landing reference speed**, known as V_{REF}.

Landing Distance

The landing can be divided into two parts.

1. The airborne section and
2. The ground run or landing roll.

The first part, the airborne section starts from the landing screen height of 50 feet and ends when the aeroplanes main wheels touch the landing surface. The airborne section is usually given as being about 1000ft in length.

The second part of the landing is the ground run, ground roll or landing roll. Once the main wheels have settled onto the landing surface **reverse thrust and lift spoilers** can be activated and as the speed decreases further, the **nose wheel will then settle** onto the landing surface.

The combined length of the 'airborne section' and the 'ground run or 'landing roll' is known as the **"landing distance required".**

LANDING DISTANCE AVAILABLE (LDA)

*In short, the landing distance available is the length of runway from one threshold to another.

LANDING DISTANCE FORMULA

$$d = V\ Square/2a$$

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THE V SPEEDS

V_{EF}: The CAS at which the critical engine is assumed to fail. It is never less than V_{MCG}.

V_1 is the speed at which, if the failure of the critical engine was recognised, there is sufficient distance remaining to either abandon the take-off or continue the take-off.

The speed, at which the critical engine fails, so that it may be recognised at V_1, is called V_{EF}.

V_1 – DECISION SPEED

V_1 is defined as being the maximum speed (V_{STOP}) at which the pilot must take the first action in order to stop the aeroplane within the remaining accelerate stop distance. **V_1** is also the minimum speed (V_{GO}) following engine failure that the pilot is able to continue the take-off within the remaining take-off distance.

V_{GO} is the lowest decision speed from which a continued take-off is possible within the take-off distance available.

V_{STOP} is the highest decision speed from which the aeroplane can stop within the accelerate-stop distance available.

These two speeds are the extremes of V_1.

There are some rules about the speed for V_1. It states that V_1:

- may not be less than V_{EF} plus the speed gained with the critical engine inoperative for the time between engine failure and the point at which the pilot applies the first means of retardation.
- must not exceed V_R
- must not exceed V_{MBE}
- must not be less than V_{MCG}

If the engine were to fail before V_1, then the decision would be to abort the take-off. The reason is because, with only one engine operating, there would be insufficient take-off distance left to accelerate the aeroplane to the screen height.

If the engine were to fail after V_1, the decision is to continue the take-off.

*FACTORS AFFECTING V_1

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In essence, whatever factors change either the accelerate stop distance required or the take-of distance required, will affect V_1.

1. **MASS:** increasing mass increases V_1.
2. **CONFIGURATION:** increasing the flap angle, decreases V_1.
3. **DENSITY:** as density falls, V_1 increases.
4. **SLOPE:** downslope reduces V_1 and upslope increases V_1.
5. **WIND:** tailwinds reduce V_1, and headwinds increase V_1.

V_{MCG} - Ground Minimum Control Speed

V_{MCG} is the minimum speed on the ground at which the take-off can be safely continued, when the critical engine suddenly becomes in-operative with the remaining engines at take-off thrust.

When the aeroplane is on the ground, you cannot use the ailerons to control the yaw otherwise you might bank the wing into the ground. Therefore the only available aerodynamic surface left to control the asymmetric yaw is the **rudder**.

The only factor that controls the value of V_{MCG} is thrust, and since take-off thrust is more or less constant, then the only variable on the amount of take-off thrust generated is **air density**.

As density increases, V_{MCG} increases.

V_{MCA} is the minimum control speed in the air. The factors that affect V_{MCA} are the same as for V_{MCG}.

V_{MCA} / V_{MC} - Air Minimum Control Speed is the minimum flight speed at which the aeroplane is controllable, with a maximum of 5° bank, when the critical engine suddenly becomes inoperative with the remaining engines at take-off thrust.

V_{MCL} - Landing Minimum Control Speed: the minimum control speed during landing approach. The minimum speed with a wing engine inoperative where it is possible to decrease thrust to idle or increase thrust to maximum take-off without encountering dangerous flight characteristics.

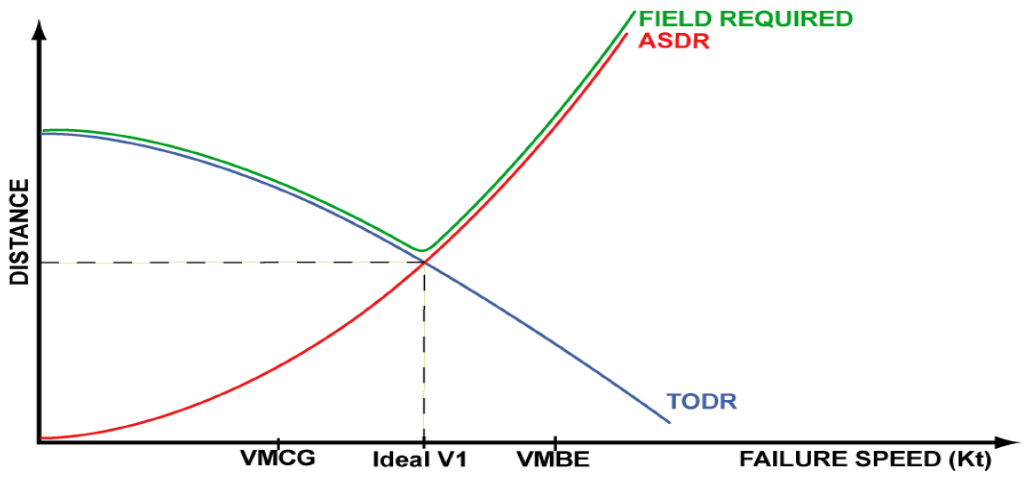
V_{MBE} - Maximum Brake Energy Speed

There are two particular speeds that can influence V_1. One of them is V_{MCG}, the other is V_{MBE}.

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V_{MBE} is the **maximum brake energy** speed and it represents the maximum speed on the ground from which an aeroplane can safely stop within the energy capabilities of the brakes.

The heavier mass reduces V_{MBE}. The variables that affect V_{MBE} are pressure altitude, ambient air temperature, mass, slope and wind.



V_1 cannot be allowed to be less than V_{MCG} because engine failure below V_{MCG} means the aeroplane is uncontrollable and the definition on V_1 is that the take-off can be continued following engine failure.

The rule also stated that V_1 must not be greater than V_{MBE} because at V_1 the aeroplane must be able to stop or continue the take-off, but above the V_{MBE} it is impossible to bring the aeroplane safely to a stop.

V_{MU} - Minimum Un-stick Speed

The speed V_{MU} is defined as the minimum un-stick speed.

V_{MU} is slowest CAS at which the aeroplane can safely lift-off the ground, and continue the take-off.

The actual speed the aeroplane will lift off, in operational flights, is called **V_{LOF}.**

V_{LOF} - Lift-off Speed

V_{LOF} means the lift-off speed.

V_{LOF} is the CAS at which the aeroplane first becomes airborne (when the main wheels have left the runway).

V_{LOF} should be faster than the minimum unstuck speed V_{MU}.

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V_R - Rotation Speed

Rotation speed, V_R, is the speed at which the pilot initiates action to raise the nose gear off the ground, with the intention of becoming airborne.

VR may not be less than

- V_1
- 1.05 V_{MC}
- a speed such that V_2 may be attained before 35ft.

Increasing mass increases V_R.
Increasing the flap angle, decreases V_R.
As density decrease, V_R increases.

Effects of early and over-rotation:

If the aircraft is rotated to the correct attitude but at too low a speed, lift off will not occur until the normal V_{LOF}, but there will be higher drag during the increased time in the rotated attitude, **giving increased distance to lift-off**.

V_2 - Take-Off Safety Speed

The speed V_2 is called the take-off safety speed.

V_2 is the **target speed** to be attained with one engine inoperative. In other words, V_2 must be reached at or prior to the screen height.

In order for V_2 to be called a safe speed it must be a faster than stall speed and the minimum control speed. There is another reason why V_2 is called the take-off safety speed.

*In the event of engine failure, V_2 must be flown until the aeroplane reaches 400 ft. Therefore, the other safe feature about V_2 is that the aeroplane is able to achieve a positive climb.

V_3: The steady initial climb speed with all engines operating.

BALANCED FIELD:

A balanced field exists if the Take-off Distance is equal to the Accelerate-stop Distance. An aerodrome which has no stopway or clearway has a balanced field.

CLIMB GRADIENT LIMIT MASS

The Climb limit mass is sometimes referred to as the Weight Altitude Limit or Mass Altitude Temperature limits, abbreviated to the **WAT or MAT** limit.

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The **climb limit mass** is the maximum mass that will enable the aeroplane to achieve a certain minimum climb performance. This minimum climb performance is the most severe of the climb gradient requirements. The most severe climb gradient requirement is in fact is **2.4%.**

RUNWAY STRENGTH

The operating mass of the aircraft may be limited by runway strength considerations. **The bearing strength of a pavement is expressed by a PCN** and this is compared to the ACN.

Operation on the pavement is permissible if the ACN is less than or equal to the PCN. Because the PCN includes a safety factor, a 10% increase of ACN over PCN is generally acceptable for pavements that are in good condition and occasional use by aircraft with ACN's up to 50% greater than the PCN may be permitted.

MAXIMUM TAKE-OFF MASS

Mass determined by the **field length available**, **the climb requirement**, **the tyre speed limit**, and **the brake energy limit** will determine the maximum mass for take-off. This mass is called the **Performance Limited Mass.**

The Performance Limited Mass must be compared to maximum structural mass and the lower of the two masses is then selected as the take-off mass. This mass is known as the **Regulated Take-off Mass**.

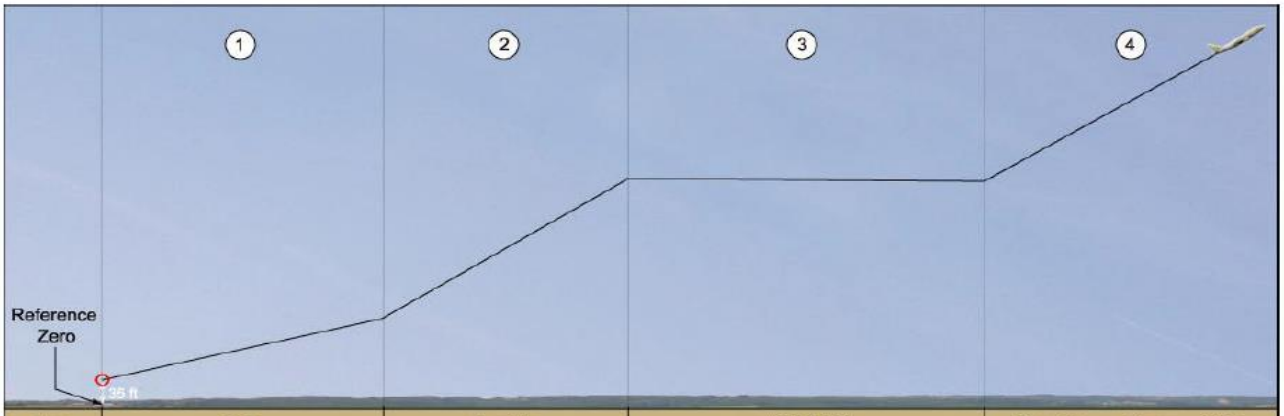
TAKE-OFF CLIMB

The take-off climb or take-off flight path extends from **35 ft above the take-off surface to 1500 ft above the take-off surface**. However, with a contaminated runway take-off, the take-off climb begins at 15 ft and not 35 ft.

SEGMENTS OF THE TAKE-OFF CLIMB

The take-off climb is generally split into four unique segments.

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Starts	35 ft	Gear Up	> 400 ft AGL	Flaps Up V_{FTO} MCT
Actions	Select Gear Up	Climb to > 400 ft AGL	Retract Flaps Accelerate to V_{FTO} Set MCT	Climb to 1500 ft AGL
Speed	V_2	V_2	V_2 V_{2F} V_{FTO}	V_{FTO}
Ends	Gear Up	> 400 ft AGL	Flaps Up V_{FTO} MCT	1500 ft AGL
Gradient	> 0 %	⩾ 2.4 %	N/A	⩾ 1.2 %
Gradient For 2 Engines	⩾ 0.3 %	⩾ 2.7 %	N/A	⩾ 1.5 %
Gradient For 4 Engines	⩾ 0.5 %	⩾ 3.0 %	N/A	⩾ 1.7 %

SEGMENT 1

The take-off flight path starts once the take-off is complete, in other words **at 35 ft** with aeroplane **at V_2 with one engine inoperative**.

The objective at this point is to climb, as expeditiously as possible.

Once the gear is up and locked then the first segment is finished. During this segment the steady gradient of climb must be positive.

SEGMENT 2

*The second segment starts **when the gear is up**.

The objective now is to retract the flaps at 400 ft, therefore the action by the pilot is simply to climb, at no less than V_2, until 400 ft is reached.

Once 400 ft is reached and flap retract can commence, segment 2 ends. The minimum gradient requirement is more severe at no less than 2.4%.

SEGMENT 3

*Segment three starts at or above 400 ft and is the **flap retraction and acceleration segment**. The aeroplane must accelerate during flap retraction from V_2 to the zero flap speed and then to the final take-off speed.

The final take-off speed is also called the **final segment speed** and is intended to be the one engine inoperative best angle of climb speed.

SEGMENT 4

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The forth segment **starts when the flaps are retracted**, the final segment speed is achieved and the thrust is set to maximum continuous thrust.

From this point the aeroplane is climbed to above 1500 ft where the take-off flight path ends.

*The climb gradient for this last stage must **not be less than 1.2%**.

*The most severe gradient requirement is in segment 2, which for a twin engine jet aeroplane is **2.4% (04/18A).**

TURNS ON THE FLIGHT PATH

1. Turns are not allowed below a height of half the wingspan or 50ft whichever is greater.
2. Up to 400ft, bank angle may not be more than 15°.
3. Above 400 ft, bank angle may not be more than 25°.

Obstacle Clearance:

An operator must ensure that the net take-off flight path must clear all obstacles by a vertical margin of at least 35 ft.

If any turn of more than 15° is required at any point in the take-off flight path, then the vertical clearance is increased to 50 ft instead of 35 ft.

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ILS

Instrument Landing System (ILS) is defined as a **precision runway approach aid** based on two radio beams which together provide pilots with both vertical and horizontal guidance during an approach to land.

The ground installation has three distinct components, namely localiser, glide path and marker beacons.

1. The **Localiser (LLZ)** transmits in the VHF band and is located about 300m from the up- wind end of the runway.
2. The **glide path (GP)** transmitter operates in the UHF band, and is frequency paired with the localiser. It is located 300m in from the threshold and about 200m from the runway edge abeam the touchdown point.
3. **Marker beacons** transmit at 75 MHZ in the VHF band. These include the outer marker (OM), the middle marker (MM) and possibly an inner marker (IM).
   ***They are provided to enable the pilot to cross check the aircraft's height against ranges and timing to the runway threshold.

ILS FREQUENCIES

1. **Localiser:** operates in the VHF band between 108 and 111.975MHz. The ILS frequencies have an odd number in the first decimal digit.
2. **Glidepath:** operates in the UHF band between 329.15 and 335MHz.
3. **Markers:** All markers transmit at 75 MHz.

DME PAIRED WITH ILS CHANNELS

1. A DME that is frequency paired with an ILS supplements or replaces the range information from markers/locators.
2. The DME ranges are zero referenced to the ILS runway threshold.
3. The DME is protected only within the ILS localiser service area up to 25,000 feet.

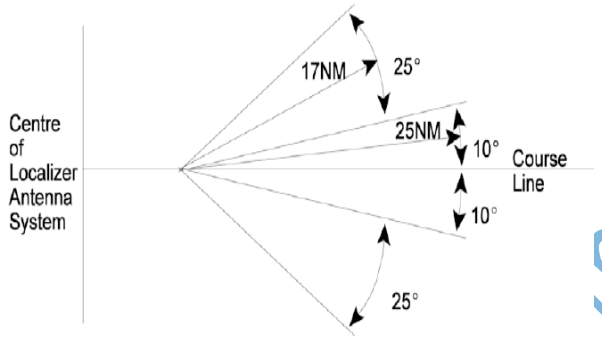
ILS IDENTIFICATION

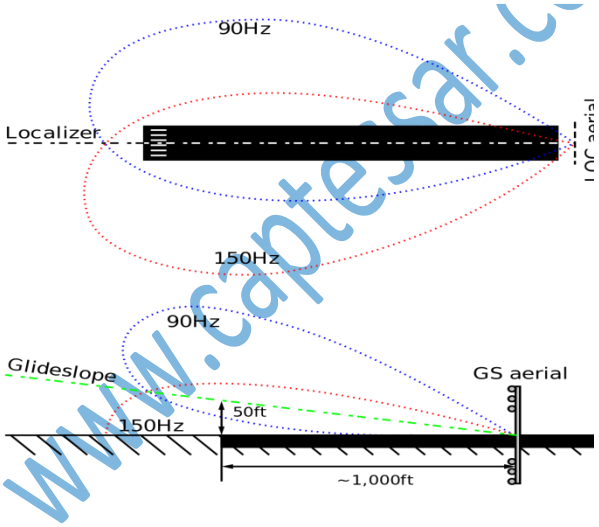
1. The Ident on the localiser transmission is a 4 letter morse signal at 7 groups/min. The first letter is usually 'I'.
2. The Identification is automatically suppressed if the ILS becomes unserviceable or is withdrawn.
3. When an ILS is **undergoing maintenance**, or is radiating for test purposes only, the identification coding will either be removed completely or replaced by a **continuous tone**.

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	Cockpit Light	Ident	Modulating Frequency	Pitch	Touchdown Range
OM	BLUE	2 dashes/sec	400 Hz	Low	6.5 – 11.1 km (3.5 – 6nm)
MM	AMBER	Alternate dots and dashes 3/sec	1,300 Hz	Medium	1050m ± 150m (3,500' ± 500')
IM	WHITE	6 dots/sec	3,000 Hz	High	75 - 450m (250' – 1,500')

ILS COVERAGE





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ILS PRINCIPLE OF OPERATION

The Localiser

1. The localiser antenna produces two overlapping lobes along the runway approach direction (QDM).
2. The lobes are transmitted on a single VHF ILS frequency. Lobes are modulated differently.
3. The depth of modulation increases away from the centre-line.
4. A DDM of zero indicates the runway centre line.

False Glideslope(s)

*These are defined as the paths of points, in the vertical plane, containing the runway centre-line at which the DDM is zero; other than that path of points forming the ILS glide path. The twin lobes are repeated due to:

1. Metallic structures situated at the transmission point, and ground reflections.
2. The height and propagation characteristics of the aerial.

*The first false glideslope occurs at approximately twice the glide path angle, 6° above ground for a standard 3° glide path. False glideslopes always occur above the true glideslope and should not constitute a danger.

ILS Reference Datum Point

The ILS reference datum point is a point at a specified height (around 50 feet) located vertically above the intersection of the runway centre-line and threshold, through which the downward extended portion of the ILS glidepath extends.

*Visual Glide Path Indicators

The approach light systems such as PAPIs give a visual indication of the glide path to the runway that would be the same as that for the ILS. The visual indications are designed for a **mean eye height (meht)** of the pilot.

Localiser Indications

Full scale deflection of the needle indicates that the aircraft is 2.5° or more left or right of the centre-line i.e. the sensitivity is 0.5° per dot.

*Glide Path Indications

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Full scale deflection indicates that the aircraft is 0.7° or more above or below the glide path. The sensitivity is 0.14° per dot.

The maximum safe deviation below the glideslope is half full-scale deflection i.e. 2.5 dots fly up.

VOR

1. The VOR produces 360 radials/tracks at 1° spacing which are aligned in relation to magnetic north at the VOR location.
2. It is practically free from static interference and is not affected by sky-waves, which enables it to be used day and night.
3. The equipment operates within the frequency range of 108 - 117.95MHz.

THE PRINCIPLE OF OPERATION

VOR bearing is obtained by phase comparison:

- An aircraft's VOR receiver measures the phase difference between two signals from the VOR transmitter:
- a 30Hz FM omni-directional, reference signal, and
- a 30Hz amplitude modulated variable phase signal.

VOR beacons operate within the VHF band between 108.0 - 117.95 MHz.

TYPES OF VOR

BVOR: A broadcast VOR which gives weather and airfield information.
DVOR: A Doppler VOR - this overcomes siting errors.
TVOR: Terminal VOR which has only low power.
VOT: At certain airfields and broadcasts a fixed omni-directional signal for a 360° test radial.

*FACTORS AFFECTING VOR BEACON ACCURACY

1. **Site error** is caused by uneven terrain such as hills and man-made structures, trees and even long grass, in the vicinity of the transmitter.
2. **Propagation error** is caused by the fact that the transmissions are further affected by terrain and distance. At considerable range from the VOR, 'bends' or 'Scalloping' can occur.
3. **VOR scalloping** is defined as an imperfection or deviation in the received VOR signal. It causes the Course Deviation Indicator to slowly or rapidly shift from side to side.

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4. **Airborne equipment errors** are caused by aircraft equipment assessing and converting the phase differences to 1° of bearing.

The above errors are aggregate errors to give a total error of + 5°.
In addition there is Pilotage Error.

*DOPPLER VOR (DVOR)

Although Doppler VOR's transmission frequencies are the same, the transmitted bearing accuracy is improved as the transmissions are less sensitive to site error.

The transmission differences are:

- The reference signal is AM.
- The variable phase directional signal is FM.
- To maintain the phase relationships which exist in conventional VOR transmissions, the rotation of the directional signal is anti-clockwise.

There are 3 main components of the VOR equipment in the aircraft, namely:

1. The aerial
2. The receiver.
3. The indicator.

RADAR: Radar stands for RAdio Detection And Ranging.

TYPES OF PULSED RADARS

1. **A Primary Radar** uses **pulses** of radio energy **reflected** from a target i.e. it uses **one frequency** throughout.
2. **A Secondary Radar** transmits pulses on one frequency, but **receives on a different frequency** i.e. the object transmits its own energy.

Primary	**Secondary**
Requires no external help	Works with the help of ground beacon
Tx & Rx on same frequency	Tx & Rx on different frequency
Uses single pulse	Uses pair of pulses
Requires high peak power	Less peak power is required
Affected by weather	Not affected by weather
Will have Tx & Rx on ground	Will have Tx & Rx on ground and a/c both
Depends on characteristics of reflecting object	Independent of reflecting object

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Examples- RADALT, GPWS, AWR SSR, DME, TCAS

RADAR APPLICATIONS

Radar has a wide range of applications as follows:

Air Traffic Control uses radar to:

- monitor aircraft in relation to each other whilst they are flying on airways, in control zones or in the airfield vicinity.
- provide radar talk-down to a given runway: SRA or a PAR
- control and monitor aircraft on ILS let-downs, or during airfield instrument approaches.
- provide information regarding weather e.g. storm clouds.

Air/Ground navigational systems use radar:

1. **SSR** provides ATC with information regarding an aircraft's callsign, altitude, speed, track history, destination and type of emergency.
2. **DME** provides a pilot with very accurate slant ranges from a ground based receiver/transmitter known as a transponder.
3. **Doppler Radar**: This is a self-contained airborne system which provides a pilot with a continuous indication of the aircraft's drift and groundspeed.

Airborne Weather Radar (AWR) is used to:

- depict the range and bearing of clouds.
- indicate areas of the heaviest precipitation and associated turbulence.
- calculate the height of cloud.
- ground map.

RADAR FREQUENCIES

Radar systems are in the VHF and above frequency bands because:

- these frequencies are free from external noise/static interference.
- the shorter wavelengths produce narrow, efficient beams for target discrimination and bearing measurement.
- the shorter wavelengths can produce shorter pulses.
- efficient reflection from an object depends upon its size in relation to the wavelength; shorter wavelengths are reflected more efficiently.

PULSE TECHNIQUE

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1. Primary, secondary, and Doppler radar systems use the pulse technique which is the transmission of radio energy in very short bursts.
2. Each burst of energy is in a pulse form of a pre-determined shape.
3. The duration of the pulse is equal to the **pulse length or width**.

THEORETICAL MAXIMUM RANGE

Maximum theoretical range is determined by the PRF.

Primary radar systems used by ATC include:

1. **Area Surveillance Radar (ASR):** These are long range radars (200 to 300nm) used for **airway surveillance** to provide range and bearing of aircraft.
2. **Terminal Area Surveillance Radar** (TAR): These are medium range radars, up to 75nm, used for controlling traffic in TMAs.
3. **Aerodrome Surveillance approach Radar:** These are short range radars providing positional information up to 25nm. Their wavelengths are 3cm or 10cm. The talkdown terminates at 2 nm from threshold.
4. **Precision Approach Radar (PAR):** The **PAR** is a **runway approach** aid available only at **military airfields.** It transmits on a frequency of 10GHz (λ 3cm) and provides the controller with very accurate azimuth and elevation information (i.e. an electronic glide-path).
   The talkdown terminates at ½ nm, or less.

5. **Airport Surface Movement RADAR (ASMR)**

This is also known as **ASMI** and is installed at major airfields to provide a very accurate radar display of the aerodrome infrastructure, vehicular traffic and aircraft that are stationary, taxi-ing, landing or taking -off.

The very high definition required by these radars is achieved by designing a radar with:

- a scanner rotation rate of 60rpm.
- frequencies of 15 to 17 GHz (SHF), 2 to 1.76cm wavelengths.
- ranges of 2.5 to 6nm in light precipitation.

AWR

Airborne weather radar (AWR) is used to provide pilots with information regarding weather ahead as well as navigation.

Information on cloud formations or terrain features is displayed on the indicator's screen as a range from the aircraft and a bearing relative to its heading.

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Primary Radar

AWR is a primary radar and both of its functions, weather detection and ground mapping, use the **echo principle** to depict **range** and the **searchlight principle** to depict **relative bearing** of the targets.

Radar Frequency

The typical frequency adopted by most commercial systems is **9375 MHz**, +/- 30 MHz as it produces the best returns from the large water droplets and wet hail found in convective clouds.

In colour weather radar systems the weather targets are colour-coded according to the intensity of the rainfall as follows:

BLACK: Very light or no returns Less than 0.7mm/hr.
GREEN: Light returns 0.7 - 4mm/hr.
YELLOW: Medium returns 4 - 12mm/hr.
RED: Strong returns Greater than 12mm/hr.
MAGENTA: Turbulence Due to rainfall intensity.

On colour systems without Magenta the RED areas may have a CYCLIC function, which causes them to alternate RED/BLACK in order to draw the pilot's attention**.**

MAPPING OPERATION

For the basic monochrome AWR with a maximum range of 150nm, **the cosecant (fan-shape)** beam is used for mapping up to about 70nm by selecting MAP. To map beyond 70 nm, the pencil beam is used.

Height of Storm Cloud

The height of a storm cloud can be ascertained by adjusting the tilt until the radar returns from it just disappear.

The height of the top of the cloud can be calculated by using the 1 in 60 Rule.

Height Ring

With the older AWR systems where the conical beam is produced by a dish antenna there is always some vertical overspill of energy which is reflected back to

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the aircraft and appears as a “height ring”, which roughly indicates the aircraft’s height.

It also indicates that the equipment is serviceable when there is no weather ahead.

Height = (TILT – BEAMWIDTH/2) x Range x 100 feet

Difference between Primary and Secondary Radar?

Primary radar relies on the reception of a reflected pulse i.e. the echo of the transmitted pulse.

Secondary radar, on the other hand, receives pulses transmitted by the target in response to interrogation pulses. Secondary surveillance radar (SSR) is one type of secondary radar system; DME is another such system.

Both Primary and Secondary Surveillance radars are used to track the progress of an aircraft.

Primary radar provides better bearing and range information of an aircraft than SSR but its biggest disadvantage is the lack of positive, individual aircraft identification. Primary radars also require higher transmitter power outputs for the two-way journey of the single pulses.

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SSR

The ground station **transmits/ interrogates** on **1030MHz** and **receives** on **1090MHz**. The aircraft **receives** on **1030MHz** and **transmits/transponds** on **1090MHz**.

MODES

The aircraft is interrogated from the ground station by a pre-determined series of pulses on the carrier frequency of 1030MHz; its transponder then transmits a coded reply on a carrier frequency of 1090MHz.

The two main modes of operation are:

- **Mode A - Identify** an aircraft
- **Mode C** - an **automatic height read-out** of an aircraft.

Special Purpose Codes

- **7700** To indicate an emergency condition, this code should be selected as soon as is practicable after declaring an emergency situation.
- **7600** To indicate a radio failure.
- **7500** To indicate unlawful interference with the planned operation of the flight, unless circumstances warrant the operation of code 7700.
- **2000** To provide recognition of an aircraft which has not received any instructions from ATC units.

MODE C should be operated with all of the above codes.

DISADVANTAGES OF SSR

1. **Garbling:** This is caused by **overlapping replies** from two or more transponders on nearly the same bearing from the ground station and within a distance of **1.7 nm from each other** measured on a line from the antenna.

2. **Fruiting:** This is interference at one interrogator caused by replies from a transponder responding to interrogations from another.

Availability of codes

Only **4096** identification codes are available in Mode A.

MODES

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1. Mode S is being introduced in order to overcome the limitations of the present modes A and C. 'S' stands for **Selective** addressing.
2. The new system has to be compatible with the existing modes A and C so that it can be used to supplement the present system.

The main features of the new **mode S** are:

1. **Availability of codes**: The system will have **over 16,700,000** discrete codes available for allocation to individual aircraft on a permanent basis. The code will be incorporated into the aircraft at manufacture and remain with it throughout its life.
2. **Data link:** The system will be supported by a ground data network and will have the ability to handle uplink/downlink data messages over the horizon. Mode S can provide ground-to- air, air-to ground and air-to-air data exchange using communications protocols.
3. **Reduction of Voice Communications:** Messages to and from an aircraft will be exchanged via the aircraft's CDU resulting in a reduction in voice communications.

GNSS

GPS is an electronic navigation system in which aircraft receivers compare signals from several of the 24 transmitters in the satellites which make up the GPS constellation. It can be received over the whole globe and its accuracy is of the order of tens of metres.

The NAVSTAR Global Positioning System (GPS) operated by the USA.
The Global Orbiting Navigation Satellite System (GLONASS) operated by Russia.

THE GPS SEGMENTS

GPS comprises three segments:

1. The Space Segment
2. The Control Segment and
3. The User Segment

THE SPACE SEGMENT

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In GPS the space segment comprises a constellation of 24 SVs in 6 orbital planes; 21 SVs are operational and the other 3 are in-space spares, to be activated to replace any failure of an operational SV.

The orbits have an average height of 10,898nm (20,180km) and have an orbital period of 12 hours. The orbital planes have an inclination of 55° and are equally spaced around the Equator.

The SVs broadcast **pseudo-random noise (PRN)** codes of one millisecond duration on two frequencies in the UHF band and a NAV and SYSTEM data message. Each SV has its own unique code.

L1 Frequency: 1,575.42 MHz transmits the C/A code repeated every millisecond, the precision (P) code repeats every seven days and the navigation and system data message at 50 Hz.
The navigation and system data message is used by both the P and C/A codes.
L2 Frequency: 1227.6 MHz transmitting the P code. The second frequency is used to determine ionospheric delays.
L3 Frequency: 1381.05 MHz has been allocated as a second frequency for non-authorised users.

Only the **C/A code** is available to civilian users. The **P code** is provided for the military and approved civilian users and foreign military users.

THE USER SEGMENT

The User Segment is all the GPS receivers using the space segment to determine position on and close to the surface of the Earth.

GPS ERRORS

All errors are at the 95% probability level.

1. **Ephemeris Errors:** These are errors in the SVs calculation of position caused by the gravitational effects and solar radiation.
2. **SV Clock Error.** As with SV ephemeris, the SV clock is checked at least every 12 hours and any error is passed to the SV to be included in the broadcast.
3. **Ionospheric propagation error:** This is the most significant of the errors in SV navigation systems. Maximum error for single frequency operation is 5 m.

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4. **Tropospheric propagation error:** Variations in pressure, temperature, density and humidity affect the speed of propagation, increased density and increased absolute humidity reduce the speed of propagation.

DIFFERENTIAL GPS (DGPS)

DGPS is a means of improving the accuracy of GPS by monitoring the integrity of the SV data and warning the user of any errors which occur. DGPS systems will provide warning of failure in the SV data and improve the accuracy of the deduced position. There are three kinds of DGPS currently in use or under development:

1. **Air based augmentation systems (ABAS)**
2. **Ground based augmentation systems (GBAS)**
3. **Satellite based augmentation systems (SBAS)**

Air Based Augmentation Systems (ABAS)

To determine, at the receiver, if any of the data from any of the SVs is in error requires the use of a fifth SV.

RAIM?

Receiver autonomous integrity monitoring (RAIM) provides integrity monitoring of GPS for aviation applications. In order for a GPS receiver to perform RAIM or fault detection (FD) function, a minimum of five visible satellites must be visible to it.

Ground Based Augmentation Systems (GBAS)

GBAS is a **local area DGPS (LADGPS)** implemented through a **local area augmentation system (LAAS).**

This system is used to provide both failure warning and enhancement of the GPS receiver position by removing ephemeris and SV clock errors and minimising ionospheric and tropospheric errors.

It is specifically established to provide precision runway approaches.

The LAAS has the potential to provide the necessary accuracy to achieve category 3C type operations.

Satellite Based Augmentation Systems (SBAS)

SBAS utilise a **wide area DGPS (WADGPS)** implemented through a **wide area augmentation system (WAAS)**. There are three systems currently under development which will cover a large area.

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The USA **WAAS,** declared operational in July 2003.

The Indian **Geo and GPS Augmented Navigation (GAGAN)** became operational in 2017.

FAIL SAFE AUTOPILOT

With any automatic system it is necessary to protect against malfunctions - in particular, runaways. This is achieved either by limiting the authority of the actuator. In this way the pilot should always be able to override the effects of a malfunction and retain control of the aircraft in the event of autopilot failure.

AUTOLAND

Requirements

To enable an aeroplane to complete an automatic landing the auto-land system requires:

- A minimum of **two independent autopilots**.
- Two independent **Radio Altimeters**.
- **Category 3 ILS** ground installation at the airport.

Auto-land Status

The number of autopilots required depends upon the auto-land status of the aircraft. These fall into two main categories:

1. **Fail – Passive/Safe**

This is defined as the ability of the system to withstand a failure without endangering passenger safety, and without producing excessive deviations in the flight path but removing its capability to complete an automatic landing.

The minimum number of autopilots required for a fail-passive capability is **two**.

2. **Fail-Operational/Active**

This is defined as the ability of a system to withstand a failure without affecting the overall functioning of the system and without causing degradation of performance beyond the limits required for automatic landing.

The system requires a minimum of **three** autopilots.

Describe a radio altimeter and how it works?

Radio altimeters provide an accurate height measurement from 2500 ft down to 0 ft for continuous-wave radar beams.

Radio Altimeter operates on FMCW technique.

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What is a fail passive autopilot/landing system?

A fail passive automatic pilot landing system is one that employs **two** digital control computer channels (duplex system). In the event of a single control channel failure, there is no significant out-of-trim condition or deviation of the flight path or attitude. However, the landing is not completed automatically, thus requiring the pilot to assume control of the aircraft to complete the landing.

What is a fail operational autopilot landing system?

A fail operational automatic pilot landing system is one that employs **three** digital control computer channels (triplex system). In the event of a single control channel failure below the alert height, the approach, flare, and landing can be completed by the remaining automatic systems. This allows the automatic landing system to work in a fail operational manner.

Q. What are the errors of Radio Altimeter?
Ans. Fixed Error: The indicator pointer moves in the steps of 5 feet which means 2½ feet discrepancy may be present anytime.
Mushing Error: Different height of Tx & Rx antenna with respect to terrain.

Q. How many different SSR codes can you have? What are SSR Modes?
Ans. 4096
Mode A & B: Used for identification of the aircraft.
Mode C: For automatic height information
Mode D: Still in experimental stage
Mode S: This is used for communication surveillance in TCAS.

Q. Why lower frequencies are used during night in HF?
Ans. That's because of the diurnal variation in the ionospheric density. If transmission is continued at night on a daytime frequency, a longer skip distance will result, leaving the receiver in the dead space. This is because at night, as we have seen, the electron density decreases; the signals travel higher in the ionosphere before refraction, and are refracted less. For these reasons, the working frequency is lowered at night.

Q. What is the significance and difference between ILS X, ILS Y, ILS Z?

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Ans. ICAO Doc 8168 states that a single letter suffix, starting with the **letter Z**, following the radio navigation aid type shall be **used if two or more procedures to the same runway cannot be distinguished** by the radio navigation aid type only.

End

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